MDRS SC-32 BAHADDURGHATTA (KOGUNDE)

Chitradurga Taluk & District — 577519

Student Notes

Level 1 — Class 6

ICT Workbook - Level 1

Name: _____

Class: _____ Section: _____

2026 · CC BY-SA 4.0 · Curated by Hoysala T, Computer Science Teacher

How to Use This Workbook

Write in pencil or pen. Complete Try It before checking answers. Use Check Yourself to test yourself. Fill in Today I Learned after every lesson. Use the Revision Page before each assessment. The Glossary is at the back.

Table of Contents

1.01 What is ICT? 1.02 History of Computers 1.03 Introduction to Internet 1.04 Web Browsers 1.05 Search Engines 1.06 URLs and Websites 1.07 Email Basics 1.08 Online Safety 1.09 GPS and Digital Maps 1.10 Cloud Computing Basics 1.11 Digital Footprint 1.12 ICT in Daily Life 2.01 Introduction to Graphics 2.02 Using Paint and Tux Paint 2.03 Color Theory for Digital Art 2.04 Drawing Tools and Techniques 2.05 Shapes and Patterns 2.06 Symmetric Designs 2.07 Image Editing Basics 2.08 Layers Introduction 2.09 Filters and Effects 2.10 Creating Digital Posters 2.11 Digital Art Projects 2.12 Graphics in Real Life 3.01 Types of Communication 3.02 Audio Recording Basics 3.03 Audacity Introduction 3.04 Editing Audio 3.05 Video Basics 3.06 Video Conferencing 3.07 Presentation Tools 3.08 Creating Presentations 3.09 Animation Basics 3.10 Digital Storytelling 3.11 Podcasting 3.12 Media Production Project 4.01 What is Data? 4.02 Number Systems 4.03 Binary Basics 4.04 Data Storage 4.05 File Formats 4.06 Spreadsheet Basics 4.07 Formulas and Functions 4.08 Charts and Graphs 4.09 Data Entry Practice 4.10 Sorting and Filtering 4.11 Data Analysis 4.12 Data in Daily Life 5.01 What is Programming? 5.02 Scratch Interface 5.03 Motion Blocks 5.04 Looks and Sound 5.05 Events and Control 5.06 Variables 5.07 Conditions 5.08 Loops 5.09 Drawing with Code 5.10 Storytelling in Scratch 5.11 Game Design Basics 5.12 Debugging 6.01 What is Software? 6.02 Operating Systems 6.03 Word Processors 6.04 Spreadsheets 6.05 Presentations 6.06 Web Browsers Revisited 6.07 Educational Software 6.08 Mobile Apps 6.09 Software Updates 6.10 Installing Software 6.11 File Management 6.12 Choosing Software

Chapter Map — Internet and ICT Environment

- 1.01 What is ICT?
- 1.02 History of Computers
- 1.03 Introduction to Internet
- 1.04 Web Browsers
- 1.05 Search Engines
- 1.06 URLs and Websites
- 1.07 Email Basics
- 1.08 Online Safety
- 1.09 GPS and Digital Maps
- 1.10 Cloud Computing Basics
- 1.11 Digital Footprint
- 1.12 ICT in Daily Life

Key terms to know:

Lesson 1.01 What is ICT?

Name: _____ Class: _____ Date: _____ / _____ / _____

I Will Be Able To...

[- I can define ICT and identify its three components: Information, Communication, Technology] [- I can explAIn the input-process-output cycle using a real-world example] [- I can classify everyday devices as information, communication, or technology tools]

Key Words

ICT	Technology that handles information and helps people communicate together
Information	Facts, numbers, or pictures that give us useful knowledge daily
Communication	Sharing ideas or messages between people using different tools
Technology	Tools and machines that make our daily work easier

Core Idea

ICT is the set of tools that help us handle information and communicate with others. Information is data we use - names, numbers, pictures, sounds.

Sketch the diagram or concept here.

Worked Example

Step 1: Brainstorm school activities done on paper: attendance registers, marks cards, library register, notice boards, letter to parents.
Step 2: Suggest digital alternatives: attendance app, spreadsheet for marks, digital library catalog, SMS or app notifications.

Try It (write your answers)

1. In your own words, what is ICT?

Answer: Write a short definition using what you learned today.

2. Give one real-life example of ICT.

Answer: Any correct example from your daily life.

Check Yourself

1. Write the full form of ICT

Answer: Information and Communication Technology

2. Give one example of input and one example of output from a computer

Answer: Input - typing on a keyboard; Output - text appears on screen

Today I Learned

Write one thing you learned today:

I still need help with:

Lesson 1.02 History of Computers

Name: _____ Class: _____ Date: _____ / _____ / _____

I Will Be Able To...

[- I can list the five generations of computers in chronological order] [- I can describe how computers changed in size, speed, and cost across generations] [- I can compare a modern smartphone with ENIAC and explAIn the differences]

Key Words

Generation	Stage showing how computers improved over many years gradually
Vacuum Tube	Glass switch used in earliest large room-sized computers
Transistor	Small switch that replaced tubes and made computers smaller
Microprocessor	Tiny chip containing the brain of a modern computer

Core Idea

Explain each generation with vivid analogies.

Sketch the diagram or concept here.

Worked Example

Step 1: Imagine future computers.
Step 2: Sketch or describe an invention: a pen that translates languages in real time, a mirror that shows the news and weather, a shirt that monitors your...

Try It (write your answers)

1. Explain: List the five generations of computers in chronological order.

Answer: Use the Core Idea and Worked Example to help.

2. Give one real-life example of History of Computers.

Answer: Any correct example from your daily life.

Check Yourself

1. List the five generations in order from oldest to newest

Answer: Check with your teacher.

2. What invention made personal computers possible?

Answer: Check with your teacher.

Today I Learned

Write one thing you learned today:

I still need help with:

Lesson 1.03 Introduction to Internet

Name: _____ Class: _____ Date: _____ / _____ / _____

I Will Be Able To...

[- I can define the internet and explAIn how it connects computers worldwide] [- I can trace the path of data from a phone to a server and back] [- I can identify two advantages and two disadvantages of the internet]

Key Words

Internet	Worldwide network connecting millions of computers to share information
Server	Powerful computer that stores websites and sends data quickly
Packet	Small piece of data travelling separately across the network
ISP	Company that provides internet connection to homes and schools

Core Idea

Draw a detailed diagram on the board. Your device connects to a router in your home using Wi-Fi.

Sketch the diagram or concept here.

Worked Example

Step 1: Discuss both scenarios in groups.
Step 2: For scenario 1: no online classes, no digital payments, no social media, bank ATMs might stop working.

Try It (write your answers)

1. In your own words, what is the internet?

Answer: Write a short definition using what you learned today.

2. Give one real-life example of Introduction to Internet.

Answer: Any correct example from your daily life.

Check Yourself

1. In your own words, what is the internet?

Answer: The internet is a global network of computers connected by cables and wireless signals

2. What are packets and why does the internet use them?

Answer: Packets are small pieces of data that travel independently and reassemble at the destination

Today I Learned

Write one thing you learned today:

I still need help with:

Lesson 1.04 Web Browsers

Name: _____ Class: _____ Date: _____ / _____ / _____

I Will Be Able To...

[- I can identify the mAIn parts of a web browser interface: address bar, back, forward, refresh, tabs, bookmarks] [- I can navigate to websites by typing URLs in the address bar] [- I can use bookmarks to save and quickly access frequently visited websites]

Key Words

Browser	Program that lets you open and view websites easily
Address Bar	Long box at top where you type website address
Tab	Small window letting you open many sites together
Bookmark	Saved link that helps you find favourite sites quickly

Core Idea

Explain each feature with demonstrations. Address bar: Type the website address here - the browser sends a request to the server at that address.

Sketch the diagram or concept here.

Worked Example

Step 1: Type each address carefully and press Enter.
Step 2: Explore each website briefly.

Try It (write your answers)

1. List or name: the mAIn parts of a web browser interface: address bar, back, forward, refresh, tabs, bookmarks.

Answer: See Key Words and Core Idea above.

2. Give one real-life example of Web Browsers.

Answer: Any correct example from your daily life.

Check Yourself

1. Type a URL and load a website,

Answer: Check with your teacher.

2. Use the back button,

Answer: Check with your teacher.

Today I Learned

Write one thing you learned today:

_____

I still need help with:

Lesson 1.05 Search Engines

Name: _____ Class: _____ Date: _____ / _____ / _____

I Will Be Able To...

[- I can explAIn how search engines find, index, and rank web pages] [- I can use three search techniques: exact phrases, exclusion, and site-specific search] [- I can evaluate whether a search result is trustworthy based on source and date]

Key Words

Search Engine	Tool that finds websites matching your typed words quickly
Keyword	Important word you type to find specific information online
Index	Huge list where search engine stores information from websites
Ranking	Ordering of search results from most useful to least

Core Idea

Explain how search engines work in three steps. Step 1 - Crawling: crawlers visit every webpage and read content, following links from page to page.

Sketch the diagram or concept here.

Worked Example

Step 1: Search with three different queries.
Step 2: Record the top result for each.

Try It (write your answers)

1. Explain: Explain how search engines find, index, and rank web pages.

Answer: Use the Core Idea and Worked Example to help.

2. Give one real-life example of Search Engines.

Answer: Any correct example from your daily life.

Check Yourself

1. What are the three steps a search engine uses?

Answer: Check with your teacher.

2. Write a search query to find information about tigers only from .gov websites

Answer: Check with your teacher.

Today I Learned

Write one thing you learned today:

I still need help with:

Lesson 1.06 URLs and Websites

Name: _____ Class: _____ Date: _____ / _____ / _____

I Will Be Able To...

[- I can identify the four mAIn parts of a URL: protocol, domAIn name, top-level domAIn, and path] [- I can distinguish between secure HTTPS and non-secure HTTP websites] [- I can classify websites as educational, commercial, government, or organizational]

Key Words

URL	Address that shows exact location of a website online
Protocol	Rule that controls how data travels between computers securely
Domain	Unique name identifying a website on the internet
HTTPS	Secure rule that encrypts data to keep information safe

Core Idea

Explain each part. Protocol: rules for data travel - http is older, https encrypts data (s = secure). Domain name: unique website name.

Sketch the diagram or concept here.

Worked Example

Step 1: Discuss safety: always check for https and lock icon.
Step 2: Complete classification: youtube.com (commercial), ncert.nic.in (educational), india.gov.in (government), redcross.org (organization).

Try It (write your answers)

1. List or name: the four mAIn parts of a URL: protocol, domAIn name, top-level domAIn, and path.

Answer: See Key Words and Core Idea above.

2. Give one real-life example of URLs and Websites.

Answer: Any correct example from your daily life.

Check Yourself

1. Label the parts of: https://www.school.edu.in/students

Answer: Check with your teacher.

2. What does the s in https mean?

Answer: Check with your teacher.

Today I Learned

Write one thing you learned today:

I still need help with:

Lesson 1.07 Email Basics

Name: _____ Class: _____ Date: _____ / _____ / _____

I Will Be Able To...

[- I can identify the parts of an emAIl address: username, @ symbol, domAIn] [- I can compose a formal emAIl with subject line, greeting, body, and closing] [- I can use CC, BCC, and Reply All appropriately]

Key Words

Email	Message sent instantly through the internet to another person
Inbox	Folder where all received emails are stored together
Subject	Short line telling what your email is about clearly
Attachment	File like photo or document sent together with email

Core Idea

Explain each part. Email address: username@domain. Subject line: short summary like a title. Greeting: Dear Sir/ Madam (formal), Hi Name (informal).

Sketch the diagram or concept here.

Worked Example

Step 1: Write three emails.
Step 2: Email 1 (formal): To principal, subject "Science Fair Permission", polite body.

Try It (write your answers)

1. List or name: the parts of an emAIl address: username, @ symbol, domAIn.

Answer: See Key Words and Core Idea above.

2. Give one real-life example of Email Basics.

Answer: Any correct example from your daily life.

Check Yourself

1. What is the most important thing you learned today?

Answer: Your own answer.

2. Give one example of how this topic is used in real life.

Answer: Any correct real-life example.

Today I Learned

Write one thing you learned today:

I still need help with:

Lesson 1.08 Online Safety

Name: _____ Class: _____ Date: _____ / _____ / _____

I Will Be Able To...

[- I can identify five common online dangers: phishing, cyberbullying, privacy leaks, inappropriate content, and addICTion] [- I can apply the STOP rule: Stop, Think, Observe, Protect] [- I can create a personal online safety checklist]

Key Words

Phishing	Fake message tricking you to share passwords or money
Cyberbullying	Using internet to hurt or scare another person repeatedly
Privacy	Keeping personal information safe from strangers online always
STOP Rule	Stop Think Observe Protect steps before clicking anything online

Core Idea

Explain the STOP rule. S - Stop: don't click, reply, or share. T - Think: is this real? Is this safe?

Sketch the diagram or concept here.

Worked Example

Step 1: Brainstorm 10 safety rules: never share passwords, don't click unknown links, tell an adult if something feels wrong, use a nickname, don't share...

Try It (write your answers)

1. List or name: five common online dangers: phishing, cyberbullying, privacy leaks, inappropriate content, and addICTion.

Answer: See Key Words and Core Idea above.

2. Give one real-life example of Online Safety.

Answer: Any correct example from your daily life.

Check Yourself

1. Email saying account will be deleted unless you click

Answer: Check with your teacher.

2. Classmate posts embarrassing photo

Answer: Check with your teacher.

Today I Learned

Write one thing you learned today:

I still need help with:

Lesson 1.09 GPS and Digital Maps

Name: _____ Class: _____ Date: _____ / _____ / _____

I Will Be Able To...

[- I can explAIn how GPS uses satellites to determine a location on Earth] [- I can identify three real-world uses of GPS: navigation, agriculture, and emergency services] [- I can read a digital map to find directions between two points]

Key Words

GPS	System using satellites to find exact location on Earth
Satellite	Machine orbiting Earth that sends signals to your phone
Coordinates	Numbers showing exact place of a location on map
Navigation	Finding best route from one place to another safely

Core Idea

Explain GPS. GPS = Global Positioning System. 31 satellites orbit Earth. Your phone receives signals from at least 4 satellites.

Sketch the diagram or concept here.

Worked Example

Step 1: Study the route.
Step 2: Identify distance and time.

Try It (write your answers)

1. Explain: Explain how GPS uses satellites to determine a location on Earth.

Answer: Use the Core Idea and Worked Example to help.

2. Give one real-life example of GPS and Digital Maps.

Answer: Any correct example from your daily life.

Check Yourself

1. What does GPS stand for?

Answer: Check with your teacher.

2. How many satellites for location?

Answer: Check with your teacher.

Today I Learned

Write one thing you learned today:

I still need help with:

Lesson 1.10 Cloud Computing Basics

Name: _____ Class: _____ Date: _____ / _____ / _____

I Will Be Able To...

[- I can define cloud computing and explAIn how data is stored on remote servers] [- I can compare local storage (USB drive) with cloud storage (Google Drive)] [- I can identify three advantages of cloud storage: access anywhere, automatic backup, sharing]

Key Words

Cloud	Servers on internet that store your files safely always
Server	Powerful computer storing many files for many users together
Upload	Sending your file from device to cloud storage
Download	Bringing a file from cloud to your own device

Core Idea

Explain cloud computing. "The cloud" is a building full of powerful computers called servers. Companies like Google, Microsoft, Amazon own them.

Sketch the diagram or concept here.

Worked Example

Step 1: Prepare arguments for/against cloud storage.
Step 2: Hold mini-debate.

Try It (write your answers)

1. In your own words, what is cloud computing?

Answer: Write a short definition using what you learned today.

2. Give one real-life example of Cloud Computing Basics.

Answer: Any correct example from your daily life.

Check Yourself

1. What is "the cloud"?

Answer: Check with your teacher.

2. Two advantages, one disadvantage of cloud storage

Answer: Check with your teacher.

Today I Learned

Write one thing you learned today:

_____

I still need help with:

Lesson 1.11 Digital Footprint

Name: _____ Class: _____ Date: _____ / _____ / _____

I Will Be Able To...

[- I can define digital footprint and distinguish between active and passive footprints] [- I can identify three examples of information that create a digital footprint] [- I can apply the Think-Before-You-Post rule to protect their online reputation]

Key Words

Digital Footprint	Trail of data you leave whenever you use internet
Active Footprint	Data you share deliberately like posting photos online
Passive Footprint	Data collected secretly like cookies tracking your browsing
Reputation	What people think about you based on your online actions

Core Idea

Explain digital footprint. Active: you choose to share - social media, comments, photos, emails. You control this.

Sketch the diagram or concept here.

Worked Example

Step 1: List online activity from past month.
Step 2: Rate as green, yellow, red.

Try It (write your answers)

1. In your own words, what is digital footprint?

Answer: Write a short definition using what you learned today.

2. Give one real-life example of Digital Footprint.

Answer: Any correct example from your daily life.

Check Yourself

1. What is a digital footprint?

Answer: Check with your teacher.

2. One example active, one passive

Answer: Check with your teacher.

Today I Learned

Write one thing you learned today:

_____

I still need help with:

_____

Lesson 1.12 ICT in Daily Life

Name: _____ Class: _____ Date: _____ / _____ / _____

I Will Be Able To...

[- I can identify how ICT is used in five sectors: education, health, agriculture, banking, communication] [- I can explAIn how ICT helps rural communities access services] [- I can design a simple ICT solution for a problem in their village]

Key Words

AI	Machines mimicking human thinking to solve problems without explicit instructions always
Model	Trained system that makes predictions based on data it learned from examples
Data	Examples used to teach AI system patterns for making decisions later
Ethics	Rules ensuring fair and safe use of AI for everyone

Core Idea

Explain ICT impact with village examples. Education: student watches IIT lectures on YouTube. Health: farmer shows symptoms to doctor via video call.

Sketch the diagram or concept here.

Worked Example

Step 1: Identify a real problem.
Step 2: Brainstorm ICT solutions.

Try It (write your answers)

1. List or name: how ICT is used in five sectors: education, health, agriculture, banking, communication.

Answer: See Key Words and Core Idea above.

2. Give one real-life example of ICT in Daily Life.

Answer: Any correct example from your daily life.

Check Yourself

1. Most useful ICT tool you learned?

Answer: Check with your teacher.

2. How has ICT changed your village?

Answer: Check with your teacher.

Today I Learned

Write one thing you learned today:

I still need help with:

Quick Revision — Internet and ICT Environment

1.01 Distribute exit tickets
1.02 (1) List the five generations in order from oldest to newest
1.03 three questions: (1) In your own words, what is the internet? (2) What are…
1.04 Ask each student to demonstrate five skills: (1) Type a URL and load a website,…
1.05 (1) What are the three steps a search engine uses? (2) Write a search query to…
1.06 Quiz: (1) Label the parts of: https://www
1.07 Give a jumbled email - put parts in correct order
1.08 three new scenarios, write what you would do using STOP
1.09 (1) What does GPS stand for? (2) How many satellites for location? (3) Two uses…
1.10 Quiz: (1) What is "the cloud"? (2) Two advantages, one disadvantage of cloud…
1.11 Exit ticket: (1) What is a digital footprint? (2) One example active, one…
1.12 Chapter reflection: (1) Most useful ICT tool you learned? (2) How has ICT…

Key terms: ICT, Information, Communication, Technology, Generation, Vacuum Tube, Transistor, Microprocessor, Internet, Server

I still need help with:

Chapter Map — Graphics in ICT

- 2.01 Introduction to Graphics
- 2.02 Using Paint and Tux Paint
- 2.03 Color Theory for Digital Art
- 2.04 Drawing Tools and Techniques
- 2.05 Shapes and Patterns
- 2.06 Symmetric Designs
- 2.07 Image Editing Basics
- 2.08 Layers Introduction
- 2.09 Filters and Effects
- 2.10 Creating Digital Posters
- 2.11 Digital Art Projects
- 2.12 Graphics in Real Life

Key terms to know:

Lesson 2.01 Introduction to Graphics

Name: _____ Class: _____ Date: _____ / _____ / _____

I Will Be Able To...

[- I can define graphics and identify different types of computer images] [- I can distinguish between raster and vector graphics with real-world examples] [- I can explAIn how pixels form the building blocks of digital images]

Key Words

Graphics	Pictures or images created and edited on a computer
Canvas	Digital workspace where you draw or paint pictures freely
Tool	Option like brush or eraser that helps you create art
Palette	Set of colours you choose from to paint digitally

Core Idea

Define graphics as visual representations created or manipulated by computers.

Sketch the diagram or concept here.

Worked Example

Step 1: Students walk around the school with their notebooks, sketching and classifying each graphic they find.
Step 2: They present their findings to a partner and discuss disagreements.

Try It (write your answers)

1. In your own words, what is graphics?

Answer: Write a short definition using what you learned today.

2. Give one real-life example of Introduction to Graphics.

Answer: Any correct example from your daily life.

Check Yourself

1. What is the most important thing you learned today?

Answer: Your own answer.

2. Give one example of how this topic is used in real life.

Answer: Any correct real-life example.

Today I Learned

Write one thing you learned today:

I still need help with:

Lesson 2.02 Using Paint and Tux Paint

Name: _____ Class: _____ Date: _____ / _____ / _____

I Will Be Able To...

[- I can navigate the Tux PAInt interface and identify all major toolbars] [- I can use at least five different drawing tools to create a simple pICTure] [- I can save and reopen a drawing file in the correct format]

Key Words

Paint	Simple program for drawing and colouring pictures digitally
Brush	Tool that draws lines with chosen colour and thickness
Eraser	Tool that removes mistakes from your drawing cleanly
Fill	Colouring a closed shape quickly with one click

Core Idea

Walk through the Tux Paint interface systematically: the left toolbar for drawing tools, the bottom selector for tool options, the right panel for sta

Sketch the diagram or concept here.

Worked Example

Step 1: Students work on their drawings with focus.
Step 2: They help each other remember tool locations and share discoveries about color combinations.

Try It (write your answers)

1. List or name: Navigate the Tux PAInt interface and all major toolbars.

Answer: See Key Words and Core Idea above.

2. Give one real-life example of Using Paint and Tux Paint.

Answer: Any correct example from your daily life.

Check Yourself

1. What is the most important thing you learned today?

Answer: Your own answer.

2. Give one example of how this topic is used in real life.

Answer: Any correct real-life example.

Today I Learned

Write one thing you learned today:

I still need help with:

Lesson 2.03 Color Theory for Digital Art

Name: _____ Class: _____ Date: _____ / _____ / _____

I Will Be Able To...

[- I can identify primary, secondary, and complementary colors on the digital color wheel] [- I can use the color picker tool to select and mix exact shades] [- I can apply basic color harmony principles to create a visually pleasing digital drawing]

Key Words

Primary Colour	Red blue or yellow that makes all other colours
Secondary Colour	Colour made by mixing two primary colours together
Shade	Darker version of a colour made by adding black
Contrast	Strong difference between light and dark colours together

Core Idea

Teach the color wheel: primary colors (red, blue, yellow) cannot be made from other colors.

Sketch the diagram or concept here.

Worked Example

Step 1: Students create two versions of the sunset, carefully choosing colors from the palette.
Step 2: They write a short reflection comparing the emotional feel of warm versus cool palettes.

Try It (write your answers)

1. List or name: primary, secondary, and complementary colors on the digital color wheel.

Answer: See Key Words and Core Idea above.

2. Give one real-life example of Color Theory for Digital Art.

Answer: Any correct example from your daily life.

Check Yourself

1. What is the most important thing you learned today?

Answer: Your own answer.

2. Give one example of how this topic is used in real life.

Answer: Any correct real-life example.

Today I Learned

Write one thing you learned today:

I still need help with:

Lesson 2.04 Drawing Tools and Techniques

Name: _____ Class: _____ Date: _____ / _____ / _____

I Will Be Able To...

[- I can use the pencil, brush, spray, and line tools with control and precision] [- I can apply different brush sizes and opacity settings to create depth in a drawing] [- I can combine multiple drawing techniques to complete a structured composition]

Key Words

Pencil	Thin tool for drawing light outlines and sketches precisely
Brush	Thick tool for painting broad strokes with colour smoothly
Shape Tool	Quick way to draw perfect circles squares and lines
Colour Picker	Tool that copies any colour already on your canvas

Core Idea

Demonstrate each tool in detail on the projector: pencil for thin precise lines, brush for thick painterly strokes, spray can for textured areas, line

Sketch the diagram or concept here.

Worked Example

Step 1: Students work carefully on the campus map, switching tools as needed.
Step 2: They compare techniques with neighbors and share tips about brush sizes that work well for different elements.

Try It (write your answers)

1. Explain: Use the pencil, brush, spray, and line tools with control and precision.

Answer: Use the Core Idea and Worked Example to help.

2. Give one real-life example of Drawing Tools and Techniques.

Answer: Any correct example from your daily life.

Check Yourself

1. What is the most important thing you learned today?

Answer: Your own answer.

2. Give one example of how this topic is used in real life.

Answer: Any correct real-life example.

Today I Learned

Write one thing you learned today:

I still need help with:

Lesson 2.05 Shapes and Patterns

Name: _____ Class: _____ Date: _____ / _____ / _____

I Will Be Able To...

[- I can draw geometric shapes accurately using rectangle, circle, and polygon tools] [- I can create repeating patterns by duplicating and arranging shapes] [- I can identify mathematical properties of shapes while drawing digitally]

Key Words

Shape	Closed form like circle square or triangle with clear edges
Pattern	Design that repeats same shapes again and again regularly
Symmetry	Both sides of design looking exactly same like mirror
Repeat	Using same element many times to create pattern quickly

Core Idea

Teach the shape tools: rectangle (hold Shift for perfect squares), circle (hold Shift for perfect circles), and polygon (click vertices to create side

Sketch the diagram or concept here.

Worked Example

Step 1: Students design their patterns carefully, counting and measuring to ensure symmetry.
Step 2: They rotate their designs to check that all four sides look identical.

Try It (write your answers)

1. Explain: Draw geometric shapes accurately using rectangle, circle, and polygon tools.

Answer: Use the Core Idea and Worked Example to help.

2. Give one real-life example of Shapes and Patterns.

Answer: Any correct example from your daily life.

Check Yourself

1. What is the most important thing you learned today?

Answer: Your own answer.

2. Give one example of how this topic is used in real life.

Answer: Any correct real-life example.

Today I Learned

Write one thing you learned today:

I still need help with:

Lesson 2.06 Symmetric Designs

Name: _____ Class: _____ Date: _____ / _____ / _____

I Will Be Able To...

[- I can define bilateral and rotational symmetry in digital art context] [- I can use symmetry tools or manual techniques to create mirror-image designs] [- I can apply symmetry to create a butterfly, flower, or traditional motif]

Key Words

Symmetry	Balance where one half mirrors the other half perfectly
Mirror	Copying design exactly to opposite side of central line
Axis	Central line where folding gives two identical halves evenly
Mandala	Circular symmetric design made from repeated shapes beautifully

Core Idea

Define bilateral symmetry as mirror symmetry along one axis (like a human face).

Sketch the diagram or concept here.

Worked Example

Step 1: Students carefully rotate and position each petal, checking that the spacing is equal.
Step 2: They use a protractor on screen or estimate angles to achieve six-fold rotational symmetry.

Try It (write your answers)

1. In your own words, what is bilateral?

Answer: Write a short definition using what you learned today.

2. Give one real-life example of Symmetric Designs.

Answer: Any correct example from your daily life.

Check Yourself

1. What is the most important thing you learned today?

Answer: Your own answer.

2. Give one example of how this topic is used in real life.

Answer: Any correct real-life example.

Today I Learned

Write one thing you learned today:

I still need help with:

Lesson 2.07 Image Editing Basics

Name: _____ Class: _____ Date: _____ / _____ / _____

I Will Be Able To...

[- I can open, view, and perform basic edits on a digital photograph] [- I can use crop, resize, and brightness tools to improve an image] [- I can save edited images in different formats and understand format differences]

Key Words

Crop	Cutting unwanted parts to focus on important area only
Resize	Making picture bigger or smaller without losing clarity
Rotate	Turning image to change its direction or angle slightly
Filter	Effect that changes colours or mood of photo instantly

Core Idea

Demonstrate three essential editing tools: crop to remove unwanted edges and focus on the subject, resize to change image dimensions for different use

Sketch the diagram or concept here.

Worked Example

Step 1: Students work through the editing steps systematically.
Step 2: They compare the JPEG and PNG file sizes and write a sentence explaining which format is better for photos and why.

Try It (write your answers)

1. Explain: Open, view, and perform basic edits on a digital photograph.

Answer: Use the Core Idea and Worked Example to help.

2. Give one real-life example of Image Editing Basics.

Answer: Any correct example from your daily life.

Check Yourself

1. What is the most important thing you learned today?

Answer: Your own answer.

2. Give one example of how this topic is used in real life.

Answer: Any correct real-life example.

Today I Learned

Write one thing you learned today:

I still need help with:

Lesson 2.08 Layers Introduction

Name: _____ Class: _____ Date: _____ / _____ / _____

I Will Be Able To...

[- I can explAIn what layers are and why they are useful in digital art] [- I can create, name, reorder, and toggle visibility of layers in a drawing application] [- I can build a simple scene using at least three separate layers for background, middle ground, and foreground]

Key Words

Layer	Transparent sheet stacked to build complex picture step by step
Background	Bottom layer that holds main colour or photo behind everything
Foreground	Top layer that appears in front of all other layers
Opacity	How see-through or solid a layer appears on screen

Core Idea

Define layers as transparent sheets stacked on top of each other.

Sketch the diagram or concept here.

Worked Example

Step 1: Students plan their scene on paper first, deciding which elements go on which layer.
Step 2: Then they build it digitally, carefully drawing on the correct layer and checking visibility toggles.

Try It (write your answers)

1. Explain: Explain what layers are and why they are useful in digital art.

Answer: Use the Core Idea and Worked Example to help.

2. Give one real-life example of Layers Introduction.

Answer: Any correct example from your daily life.

Check Yourself

1. What is the most important thing you learned today?

Answer: Your own answer.

2. Give one example of how this topic is used in real life.

Answer: Any correct real-life example.

Today I Learned

Write one thing you learned today:

I still need help with:

Lesson 2.09 Filters and Effects

Name: _____ Class: _____ Date: _____ / _____ / _____

I Will Be Able To...

[- I can identify common image filters such as blur, sharpen, grayscale, and invert] [- I can apply at least three different filters to a photograph and observe the results] [- I can choose appropriate filters for specific artistic or practical purposes]

Key Words

Filter	Ready-made effect that changes photo look with one click
Blur	Softening image so details look less sharp and smooth
Brightness	How light or dark the whole picture appears to eyes
Effect	Special style applied to make picture more attractive quickly

Core Idea

Categorize filters into three groups: color filters (grayscale, sepia, invert, color balance) that change colors, quality filters (blur, sharpen, nois

Sketch the diagram or concept here.

Worked Example

Step 1: Students carefully apply each filter and think about which version suits each purpose.
Step 2: They discuss with partners why different contexts need different visual styles.

Try It (write your answers)

1. List or name: common image filters such as blur, sharpen, grayscale, and invert.

Answer: See Key Words and Core Idea above.

2. Give one real-life example of Filters and Effects.

Answer: Any correct example from your daily life.

Check Yourself

1. What is the most important thing you learned today?

Answer: Your own answer.

2. Give one example of how this topic is used in real life.

Answer: Any correct real-life example.

Today I Learned

Write one thing you learned today:

I still need help with:

Lesson 2.10 Creating Digital Posters

Name: _____ Class: _____ Date: _____ / _____ / _____

I Will Be Able To...

[- I can plan a poster layout including title, images, text, and color scheme] [- I can use a combination of drawing tools, text, and imported images to create a poster] [- I can apply design principles such as alignment, contrast, and hierarchy to a poster]

Key Words

Poster	Large design combining text and images to share message
Layout	Arrangement of text and pictures to look balanced neatly
Heading	Big bold words at top that catch attention quickly
Alignment	Lining up text and images neatly along invisible lines

Core Idea

Teach three design principles: hierarchy means the most important element (title) should be largest and most prominent, alignment means elements shoul

Sketch the diagram or concept here.

Worked Example

Step 1: Students translate their paper plans to the digital canvas, carefully placing each element.
Step 2: They adjust colors and sizes to make the poster visually balanced and easy to read from a distance.

Try It (write your answers)

1. Explain: Plan a poster layout including title, images, text, and color scheme.

Answer: Use the Core Idea and Worked Example to help.

2. Give one real-life example of Creating Digital Posters.

Answer: Any correct example from your daily life.

Check Yourself

1. What is the most important thing you learned today?

Answer: Your own answer.

2. Give one example of how this topic is used in real life.

Answer: Any correct real-life example.

Today I Learned

Write one thing you learned today:

I still need help with:

Lesson 2.11 Digital Art Projects

Name: _____ Class: _____ Date: _____ / _____ / _____

I Will Be Able To...

[- I can plan and execute a complete digital art project from concept to final output] [- I can integrate multiple skills learned in previous lessons into a cohesive artwork] [- I can reflect on the creative process and evaluate their own artistic growth]

Key Words

Graphics	Pictures or images created and edited on a computer
Canvas	Digital workspace where you draw or paint pictures freely
Tool	Option like brush or eraser that helps you create art
Palette	Set of colours you choose from to paint digitally

Core Idea

Review the project workflow: plan on paper, set up the canvas and layers, build the background first, add middle elements, add foreground details, app

Sketch the diagram or concept here.

Worked Example

Step 1: Students work with focused energy on their projects.
Step 2: Some draw detailed village scenes, others create festival posters, and a few design abstract patterns.

Try It (write your answers)

1. Explain: Plan and execute a complete digital art project from concept to final output.

Answer: Use the Core Idea and Worked Example to help.

2. Give one real-life example of Digital Art Projects.

Answer: Any correct example from your daily life.

Check Yourself

1. What is the most important thing you learned today?

Answer: Your own answer.

2. Give one example of how this topic is used in real life.

Answer: Any correct real-life example.

Today I Learned

Write one thing you learned today:

I still need help with:

Lesson 2.12 Graphics in Real Life

Name: _____ Class: _____ Date: _____ / _____ / _____

I Will Be Able To...

[- I can identify real-world applications of computer graphics in their dAIly lives] [- I can explAIn how graphics are used in education, communication, and entertAInment] [- I can create a graphic that solves a real problem in their school or community]

Key Words

Graphics	Pictures or images created and edited on a computer
Canvas	Digital workspace where you draw or paint pictures freely
Tool	Option like brush or eraser that helps you create art
Palette	Set of colours you choose from to paint digitally

Core Idea

Discuss three major real-world applications: education uses graphics in textbooks, presentations, and educational videos; communication uses graphics

Sketch the diagram or concept here.

Worked Example

Step 1: Students brainstorm real problems and choose the most impactful one.
Step 2: They sketch their solution on paper and begin the digital version, applying design principles, appropriate tools, and thoughtful colors.

Try It (write your answers)

1. List or name: real-world applications of computer graphics in their dAIly lives.

Answer: See Key Words and Core Idea above.

2. Give one real-life example of Graphics in Real Life.

Answer: Any correct example from your daily life.

Check Yourself

1. What is the most important thing you learned today?

Answer: Your own answer.

2. Give one example of how this topic is used in real life.

Answer: Any correct real-life example.

Today I Learned

Write one thing you learned today:

I still need help with:

Quick Revision — Graphics in ICT

2.01 Give students a quick worksheet with three images: a scanned handwriting…
2.02 Ask students to open their saved file and demonstrate one tool to a neighbor
2.03 Students present their warm and cool sunsets to a partner
2.04 Teacher displays three student drawings side by side on the projector
2.05 Students print or save their kolam patterns and present them to the class
2.06 Students submit their symmetric designs
2.07 Students display their before and after images side by side on the screen
2.08 Students demonstrate their layered scene by toggling each layer on and off…
2.09 Students display all four filtered versions and explain why they chose each…
2.10 Print the posters and hang them on the classroom wall
2.11 At the end of the period, students save their work and write a reflection: what…
2.12 Students present their problem-solving graphic to the class, explaining the…

Key terms: Graphics, Canvas, Tool, Palette, Paint, Brush, Eraser, Fill, Primary Colour, Secondary Colour

I still need help with:

Chapter Map — Audio & Visual Communication

- 3.01 Types of Communication
- 3.02 Audio Recording Basics
- 3.03 Audacity Introduction
- 3.04 Editing Audio
- 3.05 Video Basics
- 3.06 Video Conferencing
- 3.07 Presentation Tools
- 3.08 Creating Presentations
- 3.09 Animation Basics
- 3.10 Digital Storytelling
- 3.11 Podcasting
- 3.12 Media Production Project

Key terms to know:

Lesson 3.01 Types of Communication

Name: _____ Class: _____ Date: _____ / _____ / _____

I Will Be Able To...

[- I can classify communication into verbal, non-verbal, and digital categories] [- I can give examples of each communication type from dAIly school life] [- I can explAIn how digital communication differs from traditional methods]

Key Words

Communication	Exchanging information between people using chosen medium effectively
Medium	Channel like phone or video used to send message
Broadcast	Sending same message to many people at once together
Feedback	Response showing whether receiver understood your message clearly

Core Idea

Define verbal communication as using spoken words (face to face, phone, video call).

Sketch the diagram or concept here.

Worked Example

Step 1: Groups research using their own family stories and general knowledge.
Step 2: They discover that grandparents sent letters, parents used landline phones, and they use smartphones and messaging apps.

Try It (write your answers)

1. Explain: Classify communication into verbal, non-verbal, and digital categories.

Answer: Use the Core Idea and Worked Example to help.

2. Give one real-life example of Types of Communication.

Answer: Any correct example from your daily life.

Check Yourself

1. What is the most important thing you learned today?

Answer: Your own answer.

2. Give one example of how this topic is used in real life.

Answer: Any correct real-life example.

Today I Learned

Write one thing you learned today:

_____

I still need help with:

Lesson 3.02 Audio Recording Basics

Name: _____ Class: _____ Date: _____ / _____ / _____

I Will Be Able To...

[- I can explAIn how sound is captured and stored as digital audio] [- I can use a microphone to record a clear audio clip of at least 10 seconds] [- I can identify factors that affect recording quality such as background noise and distance]

Key Words

Audio	Sound or music recorded and played through computer speakers
Record	Saving your voice or sound as a digital file
Playback	Listening again to recorded sound through speakers clearly
Microphone	Device that captures your voice and sends to computer

Core Idea

Explain that a microphone converts sound waves into electrical signals, which are then converted into digital data (numbers) that the computer stores.

Sketch the diagram or concept here.

Worked Example

Step 1: Students prepare their scripts, practice speaking, and take turns recording in a quiet corner of the lab.
Step 2: They listen to playback and re-record if needed, applying the tips they learned.

Try It (write your answers)

1. Explain: Explain how sound is captured and stored as digital audio.

Answer: Use the Core Idea and Worked Example to help.

2. Give one real-life example of Audio Recording Basics.

Answer: Any correct example from your daily life.

Check Yourself

1. What is the most important thing you learned today?

Answer: Your own answer.

2. Give one example of how this topic is used in real life.

Answer: Any correct real-life example.

Today I Learned

Write one thing you learned today:

I still need help with:

Lesson 3.03 Audacity Introduction

Name: _____ Class: _____ Date: _____ / _____ / _____

I Will Be Able To...

[- I can navigate the Audacity interface and identify the record, play, stop, and edit tools] [- I can record, play back, and trim an audio clip in Audacity] [- I can save an Audacity project and export the final audio as an MP3 file]

Key Words

Audacity	Free program for recording and editing sound files easily
Track	Single line holding one sound clip inside the project
Cut	Removing unwanted part of audio to make it shorter
Export	Saving edited sound as playable file like MP3

Core Idea

Walk through Audacity interface: the transport toolbar (record, play, pause, stop, skip), the edit toolbar (selection, zoom, envelope, draw), the trac

Sketch the diagram or concept here.

Worked Example

Step 1: Students open the provided file, use the selection tool to highlight and delete the silence, then record their own voice and paste it at the...
Step 2: They listen to the final result.

Try It (write your answers)

1. List or name: Navigate the Audacity interface and the record, play, stop, and edit tools.

Answer: See Key Words and Core Idea above.

2. Give one real-life example of Audacity Introduction.

Answer: Any correct example from your daily life.

Check Yourself

1. What is the most important thing you learned today?

Answer: Your own answer.

2. Give one example of how this topic is used in real life.

Answer: Any correct real-life example.

Today I Learned

Write one thing you learned today:

I still need help with:

Lesson 3.04 Editing Audio

Name: _____ Class: _____ Date: _____ / _____ / _____

I Will Be Able To...

[- I can perform cut, copy, paste, and delete operations on audio tracks] [- I can apply volume adjustments and fade effects to improve audio quality] [- I can combine two separate audio clips into one continuous track]

Key Words

Trim	Cutting extra silence from start or end of clip
Volume	How loud or soft the sound plays to listeners
Noise Reduction	Removing background hiss to make voice clearer instantly
Fade	Gradually increasing or decreasing sound at beginning or end

Core Idea

Demonstrate four key editing operations: cut removes selected audio and copies it to clipboard, copy duplicates selected audio without removing it, pa

Sketch the diagram or concept here.

Worked Example

Step 1: Students identify each problem area by listening carefully, then apply the appropriate editing technique.
Step 2: They play back their edited version and compare it to the original.

Try It (write your answers)

1. Explain: Perform cut, copy, paste, and delete operations on audio tracks.

Answer: Use the Core Idea and Worked Example to help.

2. Give one real-life example of Editing Audio.

Answer: Any correct example from your daily life.

Check Yourself

1. What is the most important thing you learned today?

Answer: Your own answer.

2. Give one example of how this topic is used in real life.

Answer: Any correct real-life example.

Today I Learned

Write one thing you learned today:

I still need help with:

Lesson 3.05 Video Basics

Name: _____ Class: _____ Date: _____ / _____ / _____

I Will Be Able To...

[- I can identify the mAIn components of a video file: frames, resolution, and duration] [- I can use a simple video player to play, pause, and navigate through a video clip] [- I can record a short video clip using a smartphone or webcam with steady framing]

Key Words

Video	Moving pictures made from many still images played quickly
Frame	Single still picture that is part of a longer video
Timeline	Bar showing order and length of clips in your project
Clip	Short piece of video you can move or trim separately

Core Idea

Explain that video is made of frames, which are individual images displayed in rapid succession (typically 24 to 30 frames per second).

Sketch the diagram or concept here.

Worked Example

Step 1: Students practice holding the phone steady and framing their subject before recording.
Step 2: They watch their recording and re-take if the framing is off or the video is shaky.

Try It (write your answers)

1. List or name: the mAIn components of a video file: frames, resolution, and duration.

Answer: See Key Words and Core Idea above.

2. Give one real-life example of Video Basics.

Answer: Any correct example from your daily life.

Check Yourself

1. What is the most important thing you learned today?

Answer: Your own answer.

2. Give one example of how this topic is used in real life.

Answer: Any correct real-life example.

Today I Learned

Write one thing you learned today:

_____

I still need help with:

Lesson 3.06 Video Conferencing

Name: _____ Class: _____ Date: _____ / _____ / _____

I Will Be Able To...

[- I can explAIn how video conferencing works and its benefits for remote communication] [- I can navigate a video conferencing interface including mute, camera toggle, and chat features] [- I can practice proper video conferencing etiquette for educational settings]

Key Words

Video Call	Talking face-to-face through camera and internet at the same time
Mute	Turning off microphone so others cannot hear you temporarily
Share Screen	Showing your computer screen to others during a call
Bandwidth	Amount of data internet can carry each second quickly

Core Idea

Explain that video conferencing uses the internet to send both audio and video between devices in real time.

Sketch the diagram or concept here.

Worked Example

Step 1: Groups prepare their presentation and practice their delivery.
Step 2: During the conference, they take turns speaking, mute when listening, use the chat for non-urgent messages, and maintain good posture.

Try It (write your answers)

1. Explain: Explain how video conferencing works and its benefits for remote communication.

Answer: Use the Core Idea and Worked Example to help.

2. Give one real-life example of Video Conferencing.

Answer: Any correct example from your daily life.

Check Yourself

1. What is the most important thing you learned today?

Answer: Your own answer.

2. Give one example of how this topic is used in real life.

Answer: Any correct real-life example.

Today I Learned

Write one thing you learned today:

I still need help with:

Lesson 3.07 Presentation Tools

Name: _____ Class: _____ Date: _____ / _____ / _____

I Will Be Able To...

[- I can identify the features of presentation software including slides, transitions, and animations] [- I can create a simple three-slide presentation with text, images, and a consistent theme] [- I can apply a slide transition and one animation effect to enhance a presentation]

Key Words

Slide	Single page inside presentation showing text and images together
Transition	Effect that moves you smoothly from one slide next
Animation	Movement added to text or pictures on a slide
Template	Ready-made design that keeps all slides looking consistent neatly

Core Idea

Introduce LibreOffice Impress as a tool for creating slide-based presentations.

Sketch the diagram or concept here.

Worked Example

Step 1: Students gather photos of the school from the shared drive and create their slides.
Step 2: They choose a theme that matches the school colors and practice adding transitions.

Try It (write your answers)

1. List or name: the features of presentation software including slides, transitions, and animations.

Answer: See Key Words and Core Idea above.

2. Give one real-life example of Presentation Tools.

Answer: Any correct example from your daily life.

Check Yourself

1. What is the most important thing you learned today?

Answer: Your own answer.

2. Give one example of how this topic is used in real life.

Answer: Any correct real-life example.

Today I Learned

Write one thing you learned today:

I still need help with:

Lesson 3.08 Creating Presentations

Name: _____ Class: _____ Date: _____ / _____ / _____

I Will Be Able To...

[- I can plan a presentation structure with an introduction, body, and conclusion] [- I can insert and format text, images, and shapes across multiple slides] [- I can deliver a presentation with clear narration and appropriate pacing]

Key Words

Slide	Single page inside presentation showing text and images together
Transition	Effect that moves you smoothly from one slide next
Animation	Movement added to text or pictures on a slide
Template	Ready-made design that keeps all slides looking consistent neatly

Core Idea

Teach the three-part structure: introduction slide grabs attention with a title and image, body slides share the main content with a mix of text and v

Sketch the diagram or concept here.

Worked Example

Step 1: Students work on their presentations, carefully selecting images and writing concise text.
Step 2: They ask peers to check font size by standing at the back of the room.

Try It (write your answers)

1. Explain: Plan a presentation structure with an introduction, body, and conclusion.

Answer: Use the Core Idea and Worked Example to help.

2. Give one real-life example of Creating Presentations.

Answer: Any correct example from your daily life.

Check Yourself

1. What is the most important thing you learned today?

Answer: Your own answer.

2. Give one example of how this topic is used in real life.

Answer: Any correct real-life example.

Today I Learned

Write one thing you learned today:

I still need help with:

Lesson 3.09 Animation Basics

Name: _____ Class: _____ Date: _____ / _____ / _____

I Will Be Able To...

[- I can explAIn how animation creates the illusion of movement through sequential images] [- I can create a simple flipbook or frame-by-frame digital animation] [- I can identify key animation concepts: frames, keyframes, and timing]

Key Words

Animation	Creating movement by showing many pictures very quickly together
Frame	One picture in sequence that creates motion when played fast
Motion	Change of position of object from one place another
Timeline	Line showing order and timing of each frame in animation

Core Idea

Define animation as the technique of displaying a sequence of images rapidly to create the illusion of movement. Each image is a frame.

Sketch the diagram or concept here.

Worked Example

Step 1: Students plan their animation by sketching all 12 frames on paper first before drawing on the sticky notes.
Step 2: They check that each frame shows a small change from the previous one.

Try It (write your answers)

1. Explain: Explain how animation creates the illusion of movement through sequential images.

Answer: Use the Core Idea and Worked Example to help.

2. Give one real-life example of Animation Basics.

Answer: Any correct example from your daily life.

Check Yourself

1. What is the most important thing you learned today?

Answer: Your own answer.

2. Give one example of how this topic is used in real life.

Answer: Any correct real-life example.

Today I Learned

Write one thing you learned today:

I still need help with:

Lesson 3.10 Digital Storytelling

Name: _____ Class: _____ Date: _____ / _____ / _____

I Will Be Able To...

[- I can structure a story with beginning, middle, and end using digital tools] [- I can combine text, images, and audio to create a digital story] [- I can select appropriate media elements to enhance narrative impact]

Key Words

Story	Sequence of events told with words pictures and sound together
Narrative	Way you organise story with beginning middle and end
Scene	Single part of story showing one moment or place
Voiceover	Recorded speech added over pictures to explain story clearly

Core Idea

Teach the digital storytelling process: write a script (text for each slide), select or create images for each scene, record narration or add backgrou

Sketch the diagram or concept here.

Worked Example

Step 1: Students write their scripts, select images from the shared drive, and record narration.
Step 2: They practice speaking clearly and at a measured pace for the recording.

Try It (write your answers)

1. Explain: Structure a story with beginning, middle, and end using digital tools.

Answer: Use the Core Idea and Worked Example to help.

2. Give one real-life example of Digital Storytelling.

Answer: Any correct example from your daily life.

Check Yourself

1. What is the most important thing you learned today?

Answer: Your own answer.

2. Give one example of how this topic is used in real life.

Answer: Any correct real-life example.

Today I Learned

Write one thing you learned today:

I still need help with:

Lesson 3.11 Podcasting

Name: _____ Class: _____ Date: _____ / _____ / _____

I Will Be Able To...

[- I can define a podcast and identify its key elements: topic, format, and episode structure] [- I can plan and record a short podcast episode on an educational topic] [- I can apply editing techniques to produce a clean, well-paced podcast segment]

Key Words

Podcast	Series of audio episodes you can listen anytime online
Episode	Single audio story that is part of larger series
Host	Person who speaks and guides the podcast for listeners
Subscribe	Following podcast to receive new episodes automatically when released

Core Idea

Explain podcast structure: introduction (welcome and topic overview), body (main content with three key points), and conclusion (summary and what to d

Sketch the diagram or concept here.

Worked Example

Step 1: Students practice their dialogue before recording.
Step 2: During recording, they maintain eye contact for natural conversation.

Try It (write your answers)

1. In your own words, what is a podcast?

Answer: Write a short definition using what you learned today.

2. Give one real-life example of Podcasting.

Answer: Any correct example from your daily life.

Check Yourself

1. What is the most important thing you learned today?

Answer: Your own answer.

2. Give one example of how this topic is used in real life.

Answer: Any correct real-life example.

Today I Learned

Write one thing you learned today:

I still need help with:

Lesson 3.12 Media Production Project

Name: _____ Class: _____ Date: _____ / _____ / _____

I Will Be Able To...

[- I can plan and execute a complete media project combining audio, visual, and presentation elements] [- I can collaborate effectively in a production team with defined roles] [- I can reflect on the entire media production process from concept to final output]

Key Words

Concept	Main idea that helps you understand the lesson topic clearly and correctly
Example	Specific case showing how the concept works in real life situations
Practice	Repeating task many times to improve skill and remember better always
Apply	Using what you learned to solve new problem or daily task usefully

Core Idea

Walk through the production workflow: Pre-production (planning, scripting, storyboarding), Production (recording, photographing, filming), and Post-pr

Sketch the diagram or concept here.

Worked Example

Step 1: Groups work in parallel: some record audio, others prepare slides, and others select images.
Step 2: They reconvene to combine their work and review the complete piece.

Try It (write your answers)

1. Explain: Plan and execute a complete media project combining audio, visual, and presentation elements.

Answer: Use the Core Idea and Worked Example to help.

2. Give one real-life example of Media Production Project.

Answer: Any correct example from your daily life.

Check Yourself

1. What is the most important thing you learned today?

Answer: Your own answer.

2. Give one example of how this topic is used in real life.

Answer: Any correct real-life example.

Today I Learned

Write one thing you learned today:

I still need help with:

Quick Revision — Audio & Visual Communication

3.01 Each group presents their timeline
3.02 Students play their recordings for a partner
3.03 Students export their completed intro clip as MP3 and play it for the class
3.04 Students play their original and edited versions back to back for a partner
3.05 Students show their videos to a partner
3.06 After the conference, each group rates the other on etiquette using a…
3.07 Students present their three slides to a partner
3.08 Two or three students present to the class
3.09 Students show their flipbooks to the class
3.10 Two or three digital stories are played for the class
3.11 Played podcasts are evaluated by the class on: content accuracy, speaking…
3.12 Each group presents their final media piece

Key terms: Communication, Medium, Broadcast, Feedback, Audio, Record, Playback, Microphone, Audacity, Track

I still need help with:

Chapter Map — Data Representation & Processing

- 4.01 What is Data?
- 4.02 Number Systems
- 4.03 Binary Basics
- 4.04 Data Storage
- 4.05 File Formats
- 4.06 Spreadsheet Basics
- 4.07 Formulas and Functions
- 4.08 Charts and Graphs
- 4.09 Data Entry Practice
- 4.10 Sorting and Filtering
- 4.11 Data Analysis
- 4.12 Data in Daily Life

Key terms to know:

Lesson 4.01 What is Data?

Name: _____ Class: _____ Date: _____ / _____ / _____

I Will Be Able To...

[- I can define data and distinguish between raw data and processed information] [- I can identify different types of data encountered in dAIly school life] [- I can explAIn why organizing data is important for making decisions]

Key Words

Data	Facts like numbers and words stored on a computer
Raw Data	Unprocessed facts collected before cleaning or organising properly
Information	Useful meaning produced after processing raw data carefully
Dataset	Collection of related data stored together in one place

Core Idea

Define data as raw facts and figures that have not been organized (like a list of numbers).

Sketch the diagram or concept here.

Worked Example

Step 1: Students walk through the school office, library, and classroom to observe data collection in action.
Step 2: They interview the librarian about how book data is organized and the clerk about attendance data.

Try It (write your answers)

1. In your own words, what is data?

Answer: Write a short definition using what you learned today.

2. Give one real-life example of Data.

Answer: Any correct example from your daily life.

Check Yourself

1. What is the most important thing you learned today?

Answer: Your own answer.

2. Give one example of how this topic is used in real life.

Answer: Any correct real-life example.

Today I Learned

Write one thing you learned today:

I still need help with:

Lesson 4.02 Number Systems

Name: _____ Class: _____ Date: _____ / _____ / _____

I Will Be Able To...

[- I can explAIn the decimal number system and why humans use base 10] [- I can describe the binary number system and its use in computers] [- I can convert simple numbers between decimal and binary]

Key Words

Number System	Method of writing numbers using specific set of digits
Decimal	Base ten system using digits zero to nine daily
Binary	Base two system using only zero and one inside computers
Place Value	Worth of a digit depending on its position in number

Core Idea

Explain that decimal (base 10) uses digits 0-9 and each position represents a power of 10 (ones, tens, hundreds).

Sketch the diagram or concept here.

Worked Example

Step 1: Students work through the conversions carefully, checking their work with a partner.
Step 2: They create creative binary birthday cards combining math and art.

Try It (write your answers)

1. Explain: Explain the decimal number system and why humans use base 10.

Answer: Use the Core Idea and Worked Example to help.

2. Give one real-life example of Number Systems.

Answer: Any correct example from your daily life.

Check Yourself

1. What is the most important thing you learned today?

Answer: Your own answer.

2. Give one example of how this topic is used in real life.

Answer: Any correct real-life example.

Today I Learned

Write one thing you learned today:

I still need help with:

Lesson 4.03 Binary Basics

Name: _____ Class: _____ Date: _____ / _____ / _____

I Will Be Able To...

[- I can perform binary addition using the rules of base-2 arithmetic] [- I can explAIn how binary addition differs from decimal addition] [- I can apply binary addition to solve simple computer-related problems]

Key Words

Binary	Counting system using only zero and one for computers
Bit	Smallest unit of data holding either zero or one
Byte	Group of eight bits that stores one character like A
Conversion	Changing number from decimal to binary or opposite method

Core Idea

Teach binary addition rules: 0+0=0, 0+1=1, 1+0=1, 1+1=10 (write 0, carry 1), 1+1+1=11 (write 1, carry 1).

Sketch the diagram or concept here.

Worked Example

Step 1: Students work through the addition and discover that the result (11000) needs 5 bits.
Step 2: They discuss what happens in a real computer when numbers overflow their storage capacity.

Try It (write your answers)

1. Explain: Perform binary addition using the rules of base-2 arithmetic.

Answer: Use the Core Idea and Worked Example to help.

2. Give one real-life example of Binary Basics.

Answer: Any correct example from your daily life.

Check Yourself

1. What is the most important thing you learned today?

Answer: Your own answer.

2. Give one example of how this topic is used in real life.

Answer: Any correct real-life example.

Today I Learned

Write one thing you learned today:

I still need help with:

Lesson 4.04 Data Storage

Name: _____ Class: _____ Date: _____ / _____ / _____

I Will Be Able To...

[- I can identify units of digital storage: bit, byte, kilobyte, megabyte, gigabyte] [- I can estimate the storage size needed for different types of files] [- I can explAIn why storage capacity matters for choosing devices and managing files]

Key Words

Storage	Place where computer keeps data for later use safely
Memory	Temporary space where computer holds data while working actively
Hard Disk	Metal disk inside computer storing files permanently for long
Capacity	Amount of data a device can hold measured in GB

Core Idea

Define storage units: 1 bit is the smallest unit (0 or 1), 8 bits make 1 byte, 1000 bytes make 1 kilobyte (KB), 1000 KB make 1 megabyte (MB), 1000 MB

Sketch the diagram or concept here.

Worked Example

Step 1: Students calculate total storage needed (90 + 20 + 150 = 260 MB), compare with available options, and recommend the most cost-effective drive.
Step 2: They justify their recommendation.

Try It (write your answers)

1. List or name: units of digital storage: bit, byte, kilobyte, megabyte, gigabyte.

Answer: See Key Words and Core Idea above.

2. Give one real-life example of Data Storage.

Answer: Any correct example from your daily life.

Check Yourself

1. What is the most important thing you learned today?

Answer: Your own answer.

2. Give one example of how this topic is used in real life.

Answer: Any correct real-life example.

Today I Learned

Write one thing you learned today:

I still need help with:

Lesson 4.05 File Formats

Name: _____ Class: _____ Date: _____ / _____ / _____

I Will Be Able To...

[- I can identify common file formats and their extensions: .txt, .jpg, .png, .mp3, .mp4, .ods, .odp] [- I can match file formats to their appropriate uses and software] [- I can explAIn why choosing the right file format matters for quality and compatibility]

Key Words

File	Single document or picture stored with a name on computer
Format	Type deciding how file stores data like JPG or MP3
Extension	Letters after dot showing file type like .jpg or .docx
Compression	Making file smaller to save space and transfer quickly

Core Idea

Categorize file formats: text files (.txt) store plain text with no formatting, image files (.jpg for photos, .png for graphics with transparency), au

Sketch the diagram or concept here.

Worked Example

Step 1: Students discuss format choices in pairs and justify their decisions.
Step 2: They debate trade-offs between quality, file size, and compatibility.

Try It (write your answers)

1. List or name: common file formats and their extensions: .txt, .jpg, .png, .mp3, .mp4, .ods, .odp.

Answer: See Key Words and Core Idea above.

2. Give one real-life example of File Formats.

Answer: Any correct example from your daily life.

Check Yourself

1. What is the most important thing you learned today?

Answer: Your own answer.

2. Give one example of how this topic is used in real life.

Answer: Any correct real-life example.

Today I Learned

Write one thing you learned today:

I still need help with:

Lesson 4.06 Spreadsheet Basics

Name: _____ Class: _____ Date: _____ / _____ / _____

I Will Be Able To...

[- I can navigate a spreadsheet interface and identify rows, columns, and cells] [- I can enter text and numerical data into specific cells in a spreadsheet] [- I can create a simple class marks sheet with headers and student data]

Key Words

Spreadsheet	Grid of rows and columns for organising numbers and data
Cell	Single box formed by one row and one column intersection
Row	Horizontal line of cells running from left to right
Column	Vertical line of cells running from top to bottom

Core Idea

Define key spreadsheet terms: a workbook is the entire file, a worksheet is one page within it, rows are horizontal (numbered 1, 2, 3), columns are ve

Sketch the diagram or concept here.

Worked Example

Step 1: Students merge cells for the title, type headers in bold, and enter five rows of student data.
Step 2: They check alignment and formatting with a partner.

Try It (write your answers)

1. List or name: Navigate a spreadsheet interface and rows, columns, and cells.

Answer: See Key Words and Core Idea above.

2. Give one real-life example of Spreadsheet Basics.

Answer: Any correct example from your daily life.

Check Yourself

1. What is the most important thing you learned today?

Answer: Your own answer.

2. Give one example of how this topic is used in real life.

Answer: Any correct real-life example.

Today I Learned

Write one thing you learned today:

I still need help with:

Lesson 4.07 Formulas and Functions

Name: _____ Class: _____ Date: _____ / _____ / _____

I Will Be Able To...

[- I can write basic spreadsheet formulas using the equals sign (=) for addition, subtraction, multiplication, and division] [- I can use the SUM function to calculate totals for a column of numbers] [- I can apply formulas to automatically calculate student total marks and percentages]

Key Words

Formula	Rule starting with equals that calculates answer from cell values
Function	Ready-made formula like SUM that does calculations quickly automatically
Reference	Address of a cell like A1 used inside formula for calculation
Result	Answer displayed after formula finishes its calculation correctly

Core Idea

Teach formula basics: every formula starts with =, use + for addition, - for subtraction, * for multiplication, and / for division.

Sketch the diagram or concept here.

Worked Example

Step 1: Students write formulas carefully, checking that cell references are correct.
Step 2: They verify their formulas by comparing the first result with a manual calculation.

Try It (write your answers)

1. Explain: Write basic spreadsheet formulas using the equals sign (=) for addition, subtraction, multiplication, and division.

Answer: Use the Core Idea and Worked Example to help.

2. Give one real-life example of Formulas and Functions.

Answer: Any correct example from your daily life.

Check Yourself

1. What is the most important thing you learned today?

Answer: Your own answer.

2. Give one example of how this topic is used in real life.

Answer: Any correct real-life example.

Today I Learned

Write one thing you learned today:

I still need help with:

Lesson 4.08 Charts and Graphs

Name: _____ Class: _____ Date: _____ / _____ / _____

I Will Be Able To...

[- I can create a bar chart from spreadsheet data to visualize comparisons] [- I can create a pie chart to show proportions of a whole] [- I can choose the appropriate chart type for different data visualization needs]

Key Words

Chart	Picture like bar or pie showing data visually for understanding
Axis	Line along side or bottom showing measurement scale clearly
Legend	Key explaining what each colour or symbol means in chart
Data Point	Single value plotted as dot or bar inside chart

Core Idea

Teach two main chart types: bar charts compare categories (like subject averages) and pie charts show parts of a whole (like how students spend their

Sketch the diagram or concept here.

Worked Example

Step 1: Students select data ranges carefully, choosing the right cells for each chart.
Step 2: They customize chart appearance and verify that the visual matches the data.

Try It (write your answers)

1. Explain: Create a bar chart from spreadsheet data to visualize comparisons.

Answer: Use the Core Idea and Worked Example to help.

2. Give one real-life example of Charts and Graphs.

Answer: Any correct example from your daily life.

Check Yourself

1. What is the most important thing you learned today?

Answer: Your own answer.

2. Give one example of how this topic is used in real life.

Answer: Any correct real-life example.

Today I Learned

Write one thing you learned today:

I still need help with:

Lesson 4.09 Data Entry Practice

Name: _____ Class: _____ Date: _____ / _____ / _____

I Will Be Able To...

[- I can enter data accurately and efficiently into spreadsheet cells] [- I can use keyboard shortcuts to speed up data entry (Tab, Enter, Arrow keys, Ctrl+Z)] [- I can validate entered data for correctness and consistency]

Key Words

Data	Facts like numbers and words stored on a computer
Raw Data	Unprocessed facts collected before cleaning or organising properly
Information	Useful meaning produced after processing raw data carefully
Dataset	Collection of related data stored together in one place

Core Idea

Teach essential keyboard shortcuts: Tab moves to the next cell to the right, Enter moves down, arrow keys navigate in any direction, Ctrl+Z undoes the

Sketch the diagram or concept here.

Worked Example

Step 1: Students work efficiently using shortcuts, checking their work as they go.
Step 2: They use Ctrl+Z to fix mistakes without losing their rhythm.

Try It (write your answers)

1. Explain: Enter data accurately and efficiently into spreadsheet cells.

Answer: Use the Core Idea and Worked Example to help.

2. Give one real-life example of Data Entry Practice.

Answer: Any correct example from your daily life.

Check Yourself

1. What is the most important thing you learned today?

Answer: Your own answer.

2. Give one example of how this topic is used in real life.

Answer: Any correct real-life example.

Today I Learned

Write one thing you learned today:

I still need help with:

Lesson 4.10 Sorting and Filtering

Name: _____ Class: _____ Date: _____ / _____ / _____

I Will Be Able To...

[- I can sort spreadsheet data in ascending and descending order by one or more columns] [- I can apply filters to dISPlay only rows that meet specific criteria] [- I can use sorting and filtering to answer questions about a data set]

Key Words

Sort	Arranging data in order like A to Z or smallest largest
Filter	Showing only rows that meet chosen condition and hiding others
Ascending	Order from smallest to largest or A to Z upward
Descending	Order from largest to smallest or Z to A downward

Core Idea

Teach sorting: select the data range, go to Data, Sort, choose the column and ascending or descending order.

Sketch the diagram or concept here.

Worked Example

Step 1: Students sort and filter, then analyze the results.
Step 2: They identify struggling students and suggest which subjects need attention, connecting data analysis to real school decisions.

Try It (write your answers)

1. Explain: Sort spreadsheet data in ascending and descending order by one or more columns.

Answer: Use the Core Idea and Worked Example to help.

2. Give one real-life example of Sorting and Filtering.

Answer: Any correct example from your daily life.

Check Yourself

1. What is the most important thing you learned today?

Answer: Your own answer.

2. Give one example of how this topic is used in real life.

Answer: Any correct real-life example.

Today I Learned

Write one thing you learned today:

I still need help with:

Lesson 4.11 Data Analysis

Name: _____ Class: _____ Date: _____ / _____ / _____

I Will Be Able To...

[- I can calculate basic statistics from spreadsheet data: mean, median, mode, minimum, and maximum] [- I can use spreadsheet functions to automate statistical calculations] [- I can interpret statistical results to draw conclusions about a data set]

Key Words

Data	Facts like numbers and words stored on a computer
Raw Data	Unprocessed facts collected before cleaning or organising properly
Information	Useful meaning produced after processing raw data carefully
Dataset	Collection of related data stored together in one place

Core Idea

Define and teach five statistical functions: AVERAGE (mean) adds all values and divides by count, MEDIAN finds the middle value when sorted, MODE find

Sketch the diagram or concept here.

Worked Example

Step 1: Students calculate statistics for all five subjects, create a comparison table, and write a paragraph explaining their findings.
Step 2: They identify patterns and anomalies in the data.

Try It (write your answers)

1. Explain: Calculate basic statistics from spreadsheet data: mean, median, mode, minimum, and maximum.

Answer: Use the Core Idea and Worked Example to help.

2. Give one real-life example of Data Analysis.

Answer: Any correct example from your daily life.

Check Yourself

1. What is the most important thing you learned today?

Answer: Your own answer.

2. Give one example of how this topic is used in real life.

Answer: Any correct real-life example.

Today I Learned

Write one thing you learned today:

I still need help with:

Lesson 4.12 Data in Daily Life

Name: _____ Class: _____ Date: _____ / _____ / _____

I Will Be Able To...

[- I can recognize how data collection and analysis affect decisions in their dAIly lives] [- I can identify data-driven systems in their school, village, and state] [- I can create a data collection project that solves a real problem in their school community]

Key Words

Data	Facts like numbers and words stored on a computer
Raw Data	Unprocessed facts collected before cleaning or organising properly
Information	Useful meaning produced after processing raw data carefully
Dataset	Collection of related data stored together in one place

Core Idea

Discuss three real-world data systems: school attendance data helps the headmaster identify students who need support, agricultural data (rainfall, cr

Sketch the diagram or concept here.

Worked Example

Step 1: Students work in groups to design their data projects.
Step 2: They create data collection forms, plan their analysis approach, and present their proposals to the class.

Try It (write your answers)

1. Explain: Recognize how data collection and analysis affect decisions in their daily lives.

Answer: Use the Core Idea and Worked Example to help.

2. Give one real-life example of Data in Daily Life.

Answer: Any correct example from your daily life.

Check Yourself

1. What is the most important thing you learned today?

Answer: Your own answer.

2. Give one example of how this topic is used in real life.

Answer: Any correct real-life example.

Today I Learned

Write one thing you learned today:

I still need help with:

Quick Revision — Data Representation & Processing

4.01 Give students a scenario: the school principal wants to know which class has…
4.02 Give students a quick quiz: convert three decimal numbers to binary and three…
4.03 Students complete five binary addition problems of increasing difficulty
4.04 Give students a storage challenge: they have a 2 GB memory card and need to fit…
4.05 Give students a scenario worksheet: someone saved a logo as JPEG and it looks…
4.06 Teacher displays a completed marks sheet on the projector
4.07 Teacher gives a new data set (10 students, 4 subjects) and asks students to…
4.08 Display student charts on the projector
4.09 Compare student data entry speed and accuracy
4.10 Give students a new data set (15 students, 4 subjects) and five questions that…
4.11 Students present their statistical analysis to the class
4.12 Each group presents their data project proposal

Key terms: Data, Raw Data, Information, Dataset, Number System, Decimal, Binary, Place Value, Bit, Byte

I still need help with:

Chapter Map — Programming

- 5.01 What is Programming?
- 5.02 Scratch Interface
- 5.03 Motion Blocks
- 5.04 Looks and Sound
- 5.05 Events and Control
- 5.06 Variables
- 5.07 Conditions
- 5.08 Loops
- 5.09 Drawing with Code
- 5.10 Storytelling in Scratch
- 5.11 Game Design Basics
- 5.12 Debugging

Key terms to know:

Lesson 5.01 What is Programming?

Name: _____ Class: _____ Date: _____ / _____ / _____

I Will Be Able To...

[- I can define programming and explAIn why computers need instructions] [- I can write a simple set of step-by-step instructions for a everyday task] [- I can identify how programming is used in devices around them]

Key Words

Programming	Writing instructions so computer performs tasks step by step
Code	Written instructions that computer can understand and follow exactly
Instruction	Single command telling computer to do one specific action
Bug	Mistake in code that makes program behave incorrectly or stop

Core Idea

Define programming as writing instructions that a computer can follow to complete a task.

Sketch the diagram or concept here.

Worked Example

Step 1: Students write their instruction sets, then swap with a partner who follows them literally.
Step 2: When the drawing does not match, they identify which instructions were unclear and revise them.

Try It (write your answers)

1. In your own words, what is programming?

Answer: Write a short definition using what you learned today.

2. Give one real-life example of Programming.

Answer: Any correct example from your daily life.

Check Yourself

1. What is the most important thing you learned today?

Answer: Your own answer.

2. Give one example of how this topic is used in real life.

Answer: Any correct real-life example.

Today I Learned

Write one thing you learned today:

I still need help with:

Lesson 5.02 Scratch Interface

Name: _____ Class: _____ Date: _____ / _____ / _____

I Will Be Able To...

[- I can navigate the Scratch interface and identify the stage, sprite panel, block palette, and script area] [- I can create a new project and add a sprite to the stage] [- I can run a simple script by connecting motion blocks]

Key Words

Scratch	Visual programming tool using blocks to create stories and games
Stage	Area where your characters move and show their actions live
Sprite	Character or object that moves and acts on the stage
Block	Colourful puzzle piece representing one programming instruction visually

Core Idea

Walk through the Scratch interface: the stage is where the action happens (480 by 360 pixels), sprites are characters on the stage, the block palette

Sketch the diagram or concept here.

Worked Example

Step 1: Students experiment with different block combinations to achieve their desired effect.
Step 2: They change backgrounds and test their scripts multiple times.

Try It (write your answers)

1. List or name: Navigate the Scratch interface and the stage, sprite panel, block palette, and script area.

Answer: See Key Words and Core Idea above.

2. Give one real-life example of Scratch Interface.

Answer: Any correct example from your daily life.

Check Yourself

1. What is the most important thing you learned today?

Answer: Your own answer.

2. Give one example of how this topic is used in real life.

Answer: Any correct real-life example.

Today I Learned

Write one thing you learned today:

I still need help with:

Lesson 5.03 Motion Blocks

Name: _____ Class: _____ Date: _____ / _____ / _____

I Will Be Able To...

[- I can use motion blocks to move sprites in specific directions and distances] [- I can apply glide and turn blocks to create smooth movement paths] [- I can combine multiple motion blocks to navigate a sprite through a maze]

Key Words

Motion	Movement of sprite from one place to another on stage
Steps	Number of small moves sprite takes in chosen direction forward
Direction	Angle showing which way sprite will face and move toward
Coordinate	Pair of numbers showing exact position on stage like x y

Core Idea

Teach the key motion blocks: move 10 steps, turn 15 degrees, go to x y, glide 1 secs to x y, point in direction, set x to and change y by.

Sketch the diagram or concept here.

Worked Example

Step 1: Students calculate the correct turn angles: 90 degrees for a square, 120 degrees for a triangle, 60 degrees for a hexagon.
Step 2: They test and refine their scripts until the shapes are accurate.

Try It (write your answers)

1. Explain: Use motion blocks to move sprites in specific directions and distances.

Answer: Use the Core Idea and Worked Example to help.

2. Give one real-life example of Motion Blocks.

Answer: Any correct example from your daily life.

Check Yourself

1. What is the most important thing you learned today?

Answer: Your own answer.

2. Give one example of how this topic is used in real life.

Answer: Any correct real-life example.

Today I Learned

Write one thing you learned today:

I still need help with:

Lesson 5.04 Looks and Sound

Name: _____ Class: _____ Date: _____ / _____ / _____

I Will Be Able To...

[- I can use Looks blocks to change sprite costumes, size, and dISPlay speech bubbles] [- I can use Sound blocks to play sounds and change volume] [- I can create a short animation combining looks and sound blocks]

Key Words

Costume	Different appearance of sprite that you can switch quickly for animation
Backdrop	Picture behind sprites that sets scene like classroom or forest
Sound	Music or voice played when program runs for effect
Effect	Change like ghost or colour that alters how sprite looks

Core Idea

Teach Looks blocks: say Hello for 2 seconds, think Hmm for 2 seconds, switch costume to, next costume, change size by 10, show and hide.

Sketch the diagram or concept here.

Worked Example

Step 1: Students plan their animation on paper first, then build it in Scratch.
Step 2: They test and adjust timing, costume changes, and sound effects until the animation flows smoothly.

Try It (write your answers)

1. Explain: Use Looks blocks to change sprite costumes, size, and display speech bubbles.

Answer: Use the Core Idea and Worked Example to help.

2. Give one real-life example of Looks and Sound.

Answer: Any correct example from your daily life.

Check Yourself

1. What is the most important thing you learned today?

Answer: Your own answer.

2. Give one example of how this topic is used in real life.

Answer: Any correct real-life example.

Today I Learned

Write one thing you learned today:

I still need help with:

Lesson 5.05 Events and Control

Name: _____ Class: _____ Date: _____ / _____ / _____

I Will Be Able To...

[- I can use event blocks to trigger scripts when specific actions occur] [- I can apply control blocks like repeat, wAIt, and if-then to manage program flow] [- I can create a program that responds to keyboard input and runs multiple scripts simultaneously]

Key Words

Event	Action like clicking flag that starts part of program running
Control	Block that decides order or repetition of steps in program
Trigger	Signal that causes sprite to begin doing something immediately afterwards
Loop	Repeating same steps many times without writing them again

Core Idea

Teach event blocks: when green flag clicked, when space key pressed, when this sprite clicked, when backdrop switches.

Sketch the diagram or concept here.

Worked Example

Step 1: Students build four separate scripts, each with its own event trigger.
Step 2: They test all four directions and adjust the wait times so the movement feels natural.

Try It (write your answers)

1. Explain: Use event blocks to trigger scripts when specific actions occur.

Answer: Use the Core Idea and Worked Example to help.

2. Give one real-life example of Events and Control.

Answer: Any correct example from your daily life.

Check Yourself

1. What is the most important thing you learned today?

Answer: Your own answer.

2. Give one example of how this topic is used in real life.

Answer: Any correct real-life example.

Today I Learned

Write one thing you learned today:

I still need help with:

Lesson 5.06 Variables

Name: _____ Class: _____ Date: _____ / _____ / _____

I Will Be Able To...

[- I can define variables as contAIners that store and change values during a program] [- I can create and use variables to track score, count, or position in a Scratch project] [- I can apply variables to build a simple score-tracking system]

Key Words

Variable	Named box storing a value that can change during program
Value	Number or word stored inside a variable for later use
Assign	Putting a value into a variable using equals sign
Update	Changing value already stored inside a variable to new value

Core Idea

Define a variable as a named container that stores a value that can change during the program.

Sketch the diagram or concept here.

Worked Example

Step 1: Students build the timer using a forever loop with wait and change blocks, and the score using a when this sprite clicked event.
Step 2: They test and adjust the timing.

Try It (write your answers)

1. In your own words, what is variables as contAIners that store?

Answer: Write a short definition using what you learned today.

2. Give one real-life example of Variables.

Answer: Any correct example from your daily life.

Check Yourself

1. What is the most important thing you learned today?

Answer: Your own answer.

2. Give one example of how this topic is used in real life.

Answer: Any correct real-life example.

Today I Learned

Write one thing you learned today:

I still need help with:

Lesson 5.07 Conditions

Name: _____ Class: _____ Date: _____ / _____ / _____

I Will Be Able To...

[- I can explAIn what a condition is and how it controls program behavior] [- I can use if-then and if-then-else blocks to make decisions in Scratch programs] [- I can create a quiz program that gives different responses based on user answers]

Key Words

Condition	Question with true or false answer deciding next step
If	Block that runs code only when condition is true otherwise
Else	Alternative path taken when if condition is false instead
Boolean	Value that is either true or false for decision making

Core Idea

Define a condition as a statement that is either true or false, and the program decides what to do based on the result.

Sketch the diagram or concept here.

Worked Example

Step 1: Students build the quiz using ask-and-wait blocks for input, if-then-else for checking answers, and variables for score tracking.
Step 2: They test all three questions thoroughly.

Try It (write your answers)

1. Explain: Explain what a condition is and how it controls program behavior.

Answer: Use the Core Idea and Worked Example to help.

2. Give one real-life example of Conditions.

Answer: Any correct example from your daily life.

Check Yourself

1. What is the most important thing you learned today?

Answer: Your own answer.

2. Give one example of how this topic is used in real life.

Answer: Any correct real-life example.

Today I Learned

Write one thing you learned today:

I still need help with:

Lesson 5.08 Loops

Name: _____ Class: _____ Date: _____ / _____ / _____

I Will Be Able To...

[- I can explAIn why loops are essential for repeating actions efficiently in programs] [- I can use repeat-until and forever loops to control repetition in Scratch] [- I can replace multiple repeated blocks with a single loop to simplify code]

Key Words

Loop	Command that repeats steps without writing them again multiple times
Repeat	Doing same action fixed number of times automatically for efficiency
Infinite Loop	Endless repetition that never stops without break condition needed
Counter	Variable that counts how many times loop has repeated so far

Core Idea

Teach three loop types: repeat (runs a set number of times), repeat-until (runs until a condition becomes true), and forever (runs continuously).

Sketch the diagram or concept here.

Worked Example

Step 1: Students build the dance loop, testing it multiple times.
Step 2: They discover that repeat-until requires a condition and adjust until the dance completes perfectly.

Try It (write your answers)

1. Explain: Explain why loops are essential for repeating actions efficiently in programs.

Answer: Use the Core Idea and Worked Example to help.

2. Give one real-life example of Loops.

Answer: Any correct example from your daily life.

Check Yourself

1. What is the most important thing you learned today?

Answer: Your own answer.

2. Give one example of how this topic is used in real life.

Answer: Any correct real-life example.

Today I Learned

Write one thing you learned today:

I still need help with:

Lesson 5.09 Drawing with Code

Name: _____ Class: _____ Date: _____ / _____ / _____

I Will Be Able To...

[- I can use the pen extension to draw shapes and patterns programmatically] [- I can control pen color, size, and position to create artistic designs] [- I can apply loops and pen blocks to generate geometric patterns automatically]

Key Words

Pencil	Thin tool for drawing light outlines and sketches precisely
Brush	Thick tool for painting broad strokes with colour smoothly
Shape Tool	Quick way to draw perfect circles squares and lines
Colour Picker	Tool that copies any colour already on your canvas

Core Idea

Teach the pen extension blocks: pen down (starts drawing), pen up (stops drawing), set pen color to, set pen size to, clear, stamp.

Sketch the diagram or concept here.

Worked Example

Step 1: Students experiment with different turn angles and step sizes, discovering that small changes produce dramatically different patterns.
Step 2: They save their favorite designs.

Try It (write your answers)

1. Explain: Use the pen extension to draw shapes and patterns programmatically.

Answer: Use the Core Idea and Worked Example to help.

2. Give one real-life example of Drawing with Code.

Answer: Any correct example from your daily life.

Check Yourself

1. What is the most important thing you learned today?

Answer: Your own answer.

2. Give one example of how this topic is used in real life.

Answer: Any correct real-life example.

Today I Learned

Write one thing you learned today:

_____

I still need help with:

Lesson 5.10 Storytelling in Scratch

Name: _____ Class: _____ Date: _____ / _____ / _____

I Will Be Able To...

[- I can plan a digital story with characters, setting, plot, and dialogue] [- I can use Scratch blocks to create an interactive narrative with multiple scenes] [- I can combine motion, looks, sound, and event blocks to bring a story to life]

Key Words

Story	Sequence of events told with words pictures and sound together
Narrative	Way you organise story with beginning middle and end
Scene	Single part of story showing one moment or place
Voiceover	Recorded speech added over pictures to explain story clearly

Core Idea

Teach the storytelling process in Scratch: create a storyboard with scenes, assign sprites to characters, write dialogue using say blocks, add backdro

Sketch the diagram or concept here.

Worked Example

Step 1: Students build their stories scene by scene, testing each transition before moving to the next.
Step 2: They coordinate multiple sprites and backdrops to create a flowing narrative.

Try It (write your answers)

1. Explain: Plan a digital story with characters, setting, plot, and dialogue.

Answer: Use the Core Idea and Worked Example to help.

2. Give one real-life example of Storytelling in Scratch.

Answer: Any correct example from your daily life.

Check Yourself

1. What is the most important thing you learned today?

Answer: Your own answer.

2. Give one example of how this topic is used in real life.

Answer: Any correct real-life example.

Today I Learned

Write one thing you learned today:

I still need help with:

Lesson 5.11 Game Design Basics

Name: _____ Class: _____ Date: _____ / _____ / _____

I Will Be Able To...

[- I can identify the key elements of a game: objective, rules, obstacles, and feedback] [- I can create a simple Scratch game with a player-controlled sprite and a goal] [- I can apply score tracking, timing, and win conditions to a complete game]

Key Words

Concept	Main idea that helps you understand the lesson topic clearly and correctly
Example	Specific case showing how the concept works in real life situations
Practice	Repeating task many times to improve skill and remember better always
Apply	Using what you learned to solve new problem or daily task usefully

Core Idea

Teach game design elements: the objective tells the player what to achieve, rules define what actions are allowed, obstacles create challenge, feedbac

Sketch the diagram or concept here.

Worked Example

Step 1: Students build their games step by step: first the player movement, then the obstacle, then the goal collection, then the score, and finally the win...
Step 2: They test and balance the difficulty.

Try It (write your answers)

1. List or name: the key elements of a game: objective, rules, obstacles, and feedback.

Answer: See Key Words and Core Idea above.

2. Give one real-life example of Game Design Basics.

Answer: Any correct example from your daily life.

Check Yourself

1. What is the most important thing you learned today?

Answer: Your own answer.

2. Give one example of how this topic is used in real life.

Answer: Any correct real-life example.

Today I Learned

Write one thing you learned today:

I still need help with:

Lesson 5.12 Debugging

Name: _____ Class: _____ Date: _____ / _____ / _____

I Will Be Able To...

[- I can define debugging and explAIn why finding and fixing errors is a normal part of programming] [- I can identify common programming errors: syntax, logic, and runtime errors] [- I can use systematic testing strategies to find and fix bugs in Scratch projects]

Key Words

Bug	Error that makes program give wrong result or stop unexpectedly
Debug	Process of finding and fixing errors in code step by step
Test	Running program to check if it works as expected correctly
Trace	Following code line by line to find where mistake happens

Core Idea

Define debugging as the process of finding and fixing errors in a program.

Sketch the diagram or concept here.

Worked Example

Step 1: Students work through their debugging checklist systematically.
Step 2: They document each bug they find: what was the symptom, what was the cause, and how they fixed it.

Try It (write your answers)

1. In your own words, what is debugging?

Answer: Write a short definition using what you learned today.

2. Give one real-life example of Debugging.

Answer: Any correct example from your daily life.

Check Yourself

1. What is the most important thing you learned today?

Answer: Your own answer.

2. Give one example of how this topic is used in real life.

Answer: Any correct real-life example.

Today I Learned

Write one thing you learned today:

I still need help with:

Quick Revision — Programming

5.01 Each pair presents their best instruction set and the drawing it produced
5.02 Students demonstrate their projects to a partner
5.03 Students display their shape-tracing scripts
5.04 Students show their animations to the class
5.05 Students demonstrate their interactive projects
5.06 Students play each other games and report the highest scores
5.07 Students play each other quizzes and verify: Does it accept correct answers?…
5.08 Students compare two versions of the same program: one with repeated blocks and…
5.09 Students display their pen drawings on the projector
5.10 Each group performs their Scratch story for the class
5.11 Students play each other games and complete a game review form rating: fun…
5.12 Students swap their fixed projects with a partner

Key terms: Programming, Code, Instruction, Bug, Scratch, Stage, Sprite, Block, Motion, Steps

I still need help with:

Chapter Map — Software Applications

- 6.01 What is Software?
- 6.02 Operating Systems
- 6.03 Word Processors
- 6.04 Spreadsheets
- 6.05 Presentations
- 6.06 Web Browsers Revisited
- 6.07 Educational Software
- 6.08 Mobile Apps
- 6.09 Software Updates
- 6.10 Installing Software
- 6.11 File Management
- 6.12 Choosing Software

Key terms to know:

Lesson 6.01 What is Software?

Name: _____ Class: _____ Date: _____ / _____ / _____

I Will Be Able To...

[- I can define software and distinguish it from hardware] [- I can identify different types of software: system software and application software] [- I can give examples of software they use dAIly and explAIn what each one does]

Key Words

Software	Set of programs that tells hardware what to do each time
Application	Program designed for specific task like writing or drawing pictures
System Software	Programs managing computer hardware and basic operations silently behind scenes
License	Permission describing how you may legally use a program without breaking rules

Core Idea

Define software as a set of instructions that tell hardware what to do.

Sketch the diagram or concept here.

Worked Example

Step 1: Students explore the school computers, opening different programs and noting their names, types, and purposes.
Step 2: They create informative posters with accurate descriptions.

Try It (write your answers)

1. In your own words, what is software?

Answer: Write a short definition using what you learned today.

2. Give one real-life example of Software.

Answer: Any correct example from your daily life.

Check Yourself

1. What is the most important thing you learned today?

Answer: Your own answer.

2. Give one example of how this topic is used in real life.

Answer: Any correct real-life example.

Today I Learned

Write one thing you learned today:

I still need help with:

Lesson 6.02 Operating Systems

Name: _____ Class: _____ Date: _____ / _____ / _____

I Will Be Able To...

[- I can explAIn what an operating system does and why computers need one] [- I can identify the mAIn functions of an operating system: managing files, programs, and hardware] [- I can compare at least two different operating systems and their features]

Key Words

Operating System	Main software managing all hardware and programs on computer together
Kernel	Core part of operating system controlling hardware directly without visible interface
Interface	Screen and buttons letting you communicate with computer easily every time
Process	Single task currently running on the computer like playing music

Core Idea

Define an operating system (OS) as the main software that manages all other software and hardware on a computer.

Sketch the diagram or concept here.

Worked Example

Step 1: Students research the two operating systems using information provided by the teacher.
Step 2: They create detailed comparison charts and write a paragraph explaining why free software is important for schools.

Try It (write your answers)

1. Explain: Explain what an operating system does and why computers need one.

Answer: Use the Core Idea and Worked Example to help.

2. Give one real-life example of Operating Systems.

Answer: Any correct example from your daily life.

Check Yourself

1. What is the most important thing you learned today?

Answer: Your own answer.

2. Give one example of how this topic is used in real life.

Answer: Any correct real-life example.

Today I Learned

Write one thing you learned today:

I still need help with:

Lesson 6.03 Word Processors

Name: _____ Class: _____ Date: _____ / _____ / _____

I Will Be Able To...

[- I can identify the mAIn features of a word processor: typing, formatting, and saving] [- I can create a formatted document with title, headings, and body text] [- I can apply basic formatting: bold, italic, font size, and alignment]

Key Words

Word Processor	Program for typing and formatting text documents beautifully and neatly
Formatting	Changing font size colour or style to improve appearance quickly
Header	Text appearing at top of every page like title automatically
Spell Check	Tool that finds and suggests fixes for spelling mistakes instantly

Core Idea

Introduce LibreOffice Writer as a word processor.

Sketch the diagram or concept here.

Worked Example

Step 1: Students write about their village, applying each formatting requirement.
Step 2: They check their document against the checklist before saving.

Try It (write your answers)

1. List or name: the mAIn features of a word processor: typing, formatting, and saving.

Answer: See Key Words and Core Idea above.

2. Give one real-life example of Word Processors.

Answer: Any correct example from your daily life.

Check Yourself

1. What is the most important thing you learned today?

Answer: Your own answer.

2. Give one example of how this topic is used in real life.

Answer: Any correct real-life example.

Today I Learned

Write one thing you learned today:

I still need help with:

Lesson 6.04 Spreadsheets

Name: _____ Class: _____ Date: _____ / _____ / _____

I Will Be Able To...

[- I can explAIn how spreadsheets organize data in rows and columns] [- I can enter data into a spreadsheet and use basic formulas for calculations] [- I can create a simple chart from spreadsheet data]

Key Words

Spreadsheet	Grid program for calculations budgets and charts with numbers efficiently
Formula	Equation that calculates result from numbers in other cells automatically
Chart	Visual picture showing numbers as bars or pies for easy understanding
Cell Range	Group of cells selected together like A1 to B5 for calculation

Core Idea

Review spreadsheet basics: rows (numbered), columns (lettered), cells (intersection).

Sketch the diagram or concept here.

Worked Example

Step 1: Students fill in realistic amounts for their household expenses, write the total formula, and create a pie chart.
Step 2: They verify the pie chart percentages add up to 100 percent.

Try It (write your answers)

1. Explain: Explain how spreadsheets organize data in rows and columns.

Answer: Use the Core Idea and Worked Example to help.

2. Give one real-life example of Spreadsheets.

Answer: Any correct example from your daily life.

Check Yourself

1. What is the most important thing you learned today?

Answer: Your own answer.

2. Give one example of how this topic is used in real life.

Answer: Any correct real-life example.

Today I Learned

Write one thing you learned today:

I still need help with:

Lesson 6.05 Presentations

Name: _____ Class: _____ Date: _____ / _____ / _____

I Will Be Able To...

[- I can identify the features of presentation software and when to use it] [- I can create a multi-slide presentation with consistent themes and transitions] [- I can deliver a presentation with clear speaking and appropriate pacing]

Key Words

Slide	Single page of presentation holding text images and animations together nicely
Theme	Set of colours and fonts giving uniform look to all slides
Transition	Effect showing how one slide moves to the next smoothly
Presenter	Person who explains slides and talks to audience during show

Core Idea

Review LibreOffice Impress features: slide layouts, themes for consistent design, transitions between slides, animations within slides, and speaker no

Sketch the diagram or concept here.

Worked Example

Step 1: Students build their presentations carefully, checking theme consistency and practicing what they will say for each slide.

Try It (write your answers)

1. List or name: the features of presentation software and when to use it.

Answer: See Key Words and Core Idea above.

2. Give one real-life example of Presentations.

Answer: Any correct example from your daily life.

Check Yourself

1. What is the most important thing you learned today?

Answer: Your own answer.

2. Give one example of how this topic is used in real life.

Answer: Any correct real-life example.

Today I Learned

Write one thing you learned today:

I still need help with:

Lesson 6.06 Web Browsers Revisited

Name: _____ Class: _____ Date: _____ / _____ / _____

I Will Be Able To...

[- I can identify advanced web browser features: bookmarks, tabs, history, and downloads] [- I can use a web browser safely and efficiently to find and save educational resources] [- I can explAIn how web browsers work to dISPlay websites from the internet]

Key Words

Browser	Program that lets you open and view websites easily
Address Bar	Long box at top where you type website address
Tab	Small window letting you open many sites together
Bookmark	Saved link that helps you find favourite sites quickly

Core Idea

Explain how web browsers work: you type a URL or click a link, the browser sends a request to a web server (a computer that stores the website), the s

Sketch the diagram or concept here.

Worked Example

Step 1: Students search carefully using educational search terms, evaluate websites for reliability, bookmark the best ones, and download appropriate images.

Try It (write your answers)

1. List or name: advanced web browser features: bookmarks, tabs, history, and downloads.

Answer: See Key Words and Core Idea above.

2. Give one real-life example of Web Browsers Revisited.

Answer: Any correct example from your daily life.

Check Yourself

1. What is the most important thing you learned today?

Answer: Your own answer.

2. Give one example of how this topic is used in real life.

Answer: Any correct real-life example.

Today I Learned

Write one thing you learned today:

I still need help with:

Lesson 6.07 Educational Software

Name: _____ Class: _____ Date: _____ / _____ / _____

I Will Be Able To...

[- I can identify different types of educational software and their learning purposes] [- I can evaluate educational software based on content quality, ease of use, and learning value] [- I can compare traditional learning methods with software-assisted learning]

Key Words

Educational App	Program designed to teach a specific subject interactively and engagingly for learning
Simulation	Model mimicking real life to practise safely without danger or cost
Quiz	Short test checking understanding of topic just learned recently in class
Tutorial	Step-by-step guide teaching you how to use a new tool

Core Idea

Categorize educational software: drill and practice (repetitive exercises like math quizzes), tutorials (step-by-step teaching like typing tutors), si

Sketch the diagram or concept here.

Worked Example

Step 1: Students write detailed reviews based on their experience.
Step 2: They include specific examples of what they learned and suggestions for improvement.

Try It (write your answers)

1. List or name: different types of educational software and their learning purposes.

Answer: See Key Words and Core Idea above.

2. Give one real-life example of Educational Software.

Answer: Any correct example from your daily life.

Check Yourself

1. What is the most important thing you learned today?

Answer: Your own answer.

2. Give one example of how this topic is used in real life.

Answer: Any correct real-life example.

Today I Learned

Write one thing you learned today:

I still need help with:

Lesson 6.08 Mobile Apps

Name: _____ Class: _____ Date: _____ / _____ / _____

I Will Be Able To...

[- I can identify the difference between mobile apps and desktop software] [- I can evaluate mobile apps for educational value and safety] [- I can understand how apps access device features like camera, location, and contacts]

Key Words

App	Small program installed on phone to perform specific task quickly
Install	Downloading and setting up app so it appears on your phone
Permission	Allowing app to use camera or location only when you agree
Update	New version fixing bugs or adding features to existing app

Core Idea

Explain that mobile apps are software designed for smartphones and tablets.

Sketch the diagram or concept here.

Worked Example

Step 1: Students analyze their own apps, questioning why each permission is needed.
Step 2: They identify apps that ask for unnecessary permissions and rate them as risky.

Try It (write your answers)

1. List or name: the difference between mobile apps and desktop software.

Answer: See Key Words and Core Idea above.

2. Give one real-life example of Mobile Apps.

Answer: Any correct example from your daily life.

Check Yourself

1. What is the most important thing you learned today?

Answer: Your own answer.

2. Give one example of how this topic is used in real life.

Answer: Any correct real-life example.

Today I Learned

Write one thing you learned today:

I still need help with:

Lesson 6.09 Software Updates

Name: _____ Class: _____ Date: _____ / _____ / _____

I Will Be Able To...

[- I can explAIn why software needs regular updates] [- I can identify the three types of updates: security patches, bug fixes, and feature additions] [- I can understand the consequences of not updating software]

Key Words

Update	New version fixing problems or adding features to software automatically for safety
Patch	Small fix correcting security hole or bug quickly without full reinstall
Version	Number showing which release of software you are using currently on device
Upgrade	Moving to better version with more powerful features than before

Core Idea

Define software updates as changes made by the developer to improve the software.

Sketch the diagram or concept here.

Worked Example

Step 1: Students research update-related incidents and create informative posters with clear explanations and eye-catching design.

Try It (write your answers)

1. Explain: Explain why software needs regular updates.

Answer: Use the Core Idea and Worked Example to help.

2. Give one real-life example of Software Updates.

Answer: Any correct example from your daily life.

Check Yourself

1. What is the most important thing you learned today?

Answer: Your own answer.

2. Give one example of how this topic is used in real life.

Answer: Any correct real-life example.

Today I Learned

Write one thing you learned today:

I still need help with:

Lesson 6.10 Installing Software

Name: _____ Class: _____ Date: _____ / _____ / _____

I Will Be Able To...

[- I can explAIn the process of installing software on a computer] [- I can identify safe sources for downloading software] [- I can understand the difference between installing and uninstalling software]

Key Words

Install	Copying program files to computer so program runs correctly afterwards for use
Installer	Helper program that guides you through installation steps simply one by one
Wizard	Series of screens helping you set up software without needing expert knowledge
Requirement	Minimum hardware or software needed before installation succeeds properly every time

Core Idea

Define installation as the process of copying software files to the computer so it can be run.

Sketch the diagram or concept here.

Worked Example

Step 1: Students write clear, detailed installation guides with numbered steps and helpful tips for each stage of the process.

Try It (write your answers)

1. Explain: Explain the process of installing software on a computer.

Answer: Use the Core Idea and Worked Example to help.

2. Give one real-life example of Installing Software.

Answer: Any correct example from your daily life.

Check Yourself

1. What is the most important thing you learned today?

Answer: Your own answer.

2. Give one example of how this topic is used in real life.

Answer: Any correct real-life example.

Today I Learned

Write one thing you learned today:

I still need help with:

Lesson 6.11 File Management

Name: _____ Class: _____ Date: _____ / _____ / _____

I Will Be Able To...

[- I can organize files and folders using a logical directory structure] [- I can rename, move, copy, and delete files safely] [- I can create a personal file organization system for schoolwork]

Key Words

Folder	Container holding many related files together like a box
Path	Address showing where file lives like folder inside folder chain
Extension	Suffix after dot indicating file type like .txt or .mp4
Backup	Extra copy kept safe in case original is lost or damaged accidentally

Core Idea

Teach file management concepts: folders are containers that organize files, the root folder is the top level, subfolders create hierarchy, file extens

Sketch the diagram or concept here.

Worked Example

Step 1: Students build their folder structures, carefully moving and renaming files.
Step 2: They write a README explaining their logic so someone else could find files in their system.

Try It (write your answers)

1. Explain: Organize files and folders using a logical directory structure.

Answer: Use the Core Idea and Worked Example to help.

2. Give one real-life example of File Management.

Answer: Any correct example from your daily life.

Check Yourself

1. What is the most important thing you learned today?

Answer: Your own answer.

2. Give one example of how this topic is used in real life.

Answer: Any correct real-life example.

Today I Learned

Write one thing you learned today:

I still need help with:

Lesson 6.12 Choosing Software

Name: _____ Class: _____ Date: _____ / _____ / _____

I Will Be Able To...

[- I can compare different software options for the same task based on features, cost, and compatibility] [- I can make informed decisions about which software to use for specific educational tasks] [- I can explAIn the benefits of free and open-source software for education]

Key Words

Software	Set of programs that tells hardware what to do each time
Application	Program designed for specific task like writing or drawing pictures
System Software	Programs managing computer hardware and basic operations silently behind scenes
License	Permission describing how you may legally use a program without breaking rules

Core Idea

Teach software selection criteria: features (does it do what you need?), cost (can the school afford it?), compatibility (does it work on our computer

Sketch the diagram or concept here.

Worked Example

Step 1: Students research and compare options for each task, building arguments for their recommendations.
Step 2: They consider the school budget and technical requirements.

Try It (write your answers)

1. Explain: Compare different software options for the same task based on features, cost, and compatibility.

Answer: Use the Core Idea and Worked Example to help.

2. Give one real-life example of Choosing Software.

Answer: Any correct example from your daily life.

Check Yourself

1. What is the most important thing you learned today?

Answer: Your own answer.

2. Give one example of how this topic is used in real life.

Answer: Any correct real-life example.

Today I Learned

Write one thing you learned today:

I still need help with:

Quick Revision — Software Applications

6.01 Students present their posters
6.02 Students present their comparison charts and explain the advantages of each…
6.03 Students open their saved documents and display them on the projector
6.04 Students present their budget spreadsheets
6.05 Three students present to the class
6.06 Students share their bookmarked websites and downloaded images
6.07 Students present their review cards
6.08 Students share their safety checklists
6.09 Students display their posters and present their key messages
6.10 Students exchange their installation guides and follow them on a practice…
6.11 Students demonstrate their file systems
6.12 Students present their recommendations

Key terms: Software, Application, System Software, License, Operating System, Kernel, Interface, Process, Word Processor, Formatting

I still need help with:

Glossary

AI	Machines mimicking human thinking to solve problems without explicit instructions always
Active Footprint	Data you share deliberately like posting photos online
Address Bar	Long box at top where you type website address
Alignment	Lining up text and images neatly along invisible lines
Animation	Movement added to text or pictures on a slide
App	Small program installed on phone to perform specific task quickly
Application	Program designed for specific task like writing or drawing pictures
Apply	Using what you learned to solve new problem or daily task usefully
Ascending	Order from smallest to largest or A to Z upward
Assign	Putting a value into a variable using equals sign
Attachment	File like photo or document sent together with email
Audacity	Free program for recording and editing sound files easily
Audio	Sound or music recorded and played through computer speakers
Axis	Central line where folding gives two identical halves evenly
Backdrop	Picture behind sprites that sets scene like classroom or forest
Background	Bottom layer that holds main colour or photo behind everything
Backup	Extra copy kept safe in case original is lost or damaged accidentally
Bandwidth	Amount of data internet can carry each second quickly
Binary	Base two system using only zero and one inside computers
Bit	Smallest unit of data holding either zero or one
Block	Colourful puzzle piece representing one programming instruction visually
Blur	Softening image so details look less sharp and smooth
Bookmark	Saved link that helps you find favourite sites quickly
Boolean	Value that is either true or false for decision making
Brightness	How light or dark the whole picture appears to eyes
Broadcast	Sending same message to many people at once together
Browser	Program that lets you open and view websites easily
Brush	Tool that draws lines with chosen colour and thickness
Bug	Mistake in code that makes program behave incorrectly or stop
Byte	Group of eight bits that stores one character like A
Canvas	Digital workspace where you draw or paint pictures freely
Capacity	Amount of data a device can hold measured in GB
Cell	Single box formed by one row and one column intersection
Cell Range	Group of cells selected together like A1 to B5 for calculation
Chart	Picture like bar or pie showing data visually for understanding
Clip	Short piece of video you can move or trim separately
Cloud	Servers on internet that store your files safely always
Code	Written instructions that computer can understand and follow exactly
Colour Picker	Tool that copies any colour already on your canvas
Column	Vertical line of cells running from top to bottom
Communication	Sharing ideas or messages between people using different tools
Compression	Making file smaller to save space and transfer quickly
Concept	Main idea that helps you understand the lesson topic clearly and correctly
Condition	Question with true or false answer deciding next step
Contrast	Strong difference between light and dark colours together
Control	Block that decides order or repetition of steps in program
Conversion	Changing number from decimal to binary or opposite method
Coordinate	Pair of numbers showing exact position on stage like x y
Coordinates	Numbers showing exact place of a location on map
Costume	Different appearance of sprite that you can switch quickly for animation
Counter	Variable that counts how many times loop has repeated so far
Crop	Cutting unwanted parts to focus on important area only
Cut	Removing unwanted part of audio to make it shorter

Cyberbullying	Using internet to hurt or scare another person repeatedly
Data	Examples used to teach AI system patterns for making decisions later
Data Point	Single value plotted as dot or bar inside chart
Dataset	Collection of related data stored together in one place
Debug	Process of finding and fixing errors in code step by step
Decimal	Base ten system using digits zero to nine daily
Descending	Order from largest to smallest or Z to A downward
Digital Footprint	Trail of data you leave whenever you use internet
Direction	Angle showing which way sprite will face and move toward
Domain	Unique name identifying a website on the internet
Download	Bringing a file from cloud to your own device
Educational App	Program designed to teach a specific subject interactively and engagingly for learning
Effect	Special style applied to make picture more attractive quickly
Else	Alternative path taken when if condition is false instead
Email	Message sent instantly through the internet to another person
Episode	Single audio story that is part of larger series
Eraser	Tool that removes mistakes from your drawing cleanly
Ethics	Rules ensuring fair and safe use of AI for everyone
Event	Action like clicking flag that starts part of program running
Example	Specific case showing how the concept works in real life situations
Export	Saving edited sound as playable file like MP3
Extension	Letters after dot showing file type like .jpg or .docx
Fade	Gradually increasing or decreasing sound at beginning or end
Feedback	Response showing whether receiver understood your message clearly
File	Single document or picture stored with a name on computer
Fill	Colouring a closed shape quickly with one click
Filter	Effect that changes colours or mood of photo instantly
Folder	Container holding many related files together like a box
Foreground	Top layer that appears in front of all other layers
Format	Type deciding how file stores data like JPG or MP3
Formatting	Changing font size colour or style to improve appearance quickly
Formula	Rule starting with equals that calculates answer from cell values
Frame	Single still picture that is part of a longer video
Function	Ready-made formula like SUM that does calculations quickly automatically
GPS	System using satellites to find exact location on Earth
Generation	Stage showing how computers improved over many years gradually
Graphics	Pictures or images created and edited on a computer
HTTPS	Secure rule that encrypts data to keep information safe
Hard Disk	Metal disk inside computer storing files permanently for long
Header	Text appearing at top of every page like title automatically
Heading	Big bold words at top that catch attention quickly
Host	Person who speaks and guides the podcast for listeners
ICT	Technology that handles information and helps people communicate together
ISP	Company that provides internet connection to homes and schools
If	Block that runs code only when condition is true otherwise
Inbox	Folder where all received emails are stored together
Index	Huge list where search engine stores information from websites
Infinite Loop	Endless repetition that never stops without break condition needed
Information	Facts, numbers, or pictures that give us useful knowledge daily
Install	Downloading and setting up app so it appears on your phone
Installer	Helper program that guides you through installation steps simply one by one
Instruction	Single command telling computer to do one specific action
Interface	Screen and buttons letting you communicate with computer easily every time
Internet	Worldwide network connecting millions of computers to share information

Kernel	Core part of operating system controlling hardware directly without visible interface
Keyword	Important word you type to find specific information online
Layer	Transparent sheet stacked to build complex picture step by step
Layout	Arrangement of text and pictures to look balanced neatly
Legend	Key explaining what each colour or symbol means in chart
License	Permission describing how you may legally use a program without breaking rules
Loop	Repeating same steps many times without writing them again
Mandala	Circular symmetric design made from repeated shapes beautifully
Medium	Channel like phone or video used to send message
Memory	Temporary space where computer holds data while working actively
Microphone	Device that captures your voice and sends to computer
Microprocessor	Tiny chip containing the brain of a modern computer
Mirror	Copying design exactly to opposite side of central line
Model	Trained system that makes predictions based on data it learned from examples
Motion	Change of position of object from one place another
Mute	Turning off microphone so others cannot hear you temporarily
Narrative	Way you organise story with beginning middle and end
Navigation	Finding best route from one place to another safely
Noise Reduction	Removing background hiss to make voice clearer instantly
Number System	Method of writing numbers using specific set of digits
Opacity	How see-through or solid a layer appears on screen
Operating System	Main software managing all hardware and programs on computer together
Packet	Small piece of data travelling separately across the network
Paint	Simple program for drawing and colouring pictures digitally
Palette	Set of colours you choose from to paint digitally
Passive Footprint	Data collected secretly like cookies tracking your browsing
Patch	Small fix correcting security hole or bug quickly without full reinstall
Path	Address showing where file lives like folder inside folder chain
Pattern	Design that repeats same shapes again and again regularly
Pencil	Thin tool for drawing light outlines and sketches precisely
Permission	Allowing app to use camera or location only when you agree
Phishing	Fake message tricking you to share passwords or money
Place Value	Worth of a digit depending on its position in number
Playback	Listening again to recorded sound through speakers clearly
Podcast	Series of audio episodes you can listen anytime online
Poster	Large design combining text and images to share message
Practice	Repeating task many times to improve skill and remember better always
Presenter	Person who explains slides and talks to audience during show
Primary Colour	Red blue or yellow that makes all other colours
Privacy	Keeping personal information safe from strangers online always
Process	Single task currently running on the computer like playing music
Programming	Writing instructions so computer performs tasks step by step
Protocol	Rule that controls how data travels between computers securely
Quiz	Short test checking understanding of topic just learned recently in class
Ranking	Ordering of search results from most useful to least
Raw Data	Unprocessed facts collected before cleaning or organising properly
Record	Saving your voice or sound as a digital file
Reference	Address of a cell like A1 used inside formula for calculation
Repeat	Using same element many times to create pattern quickly
Reputation	What people think about you based on your online actions
Requirement	Minimum hardware or software needed before installation succeeds properly every time
Resize	Making picture bigger or smaller without losing clarity
Result	Answer displayed after formula finishes its calculation correctly

Rotate	Turning image to change its direction or angle slightly
Row	Horizontal line of cells running from left to right
STOP Rule	Stop Think Observe Protect steps before clicking anything online
Satellite	Machine orbiting Earth that sends signals to your phone
Scene	Single part of story showing one moment or place
Scratch	Visual programming tool using blocks to create stories and games
Search Engine	Tool that finds websites matching your typed words quickly
Secondary Colour	Colour made by mixing two primary colours together
Server	Powerful computer that stores websites and sends data quickly
Shade	Darker version of a colour made by adding black
Shape	Closed form like circle square or triangle with clear edges
Shape Tool	Quick way to draw perfect circles squares and lines
Share Screen	Showing your computer screen to others during a call
Simulation	Model mimicking real life to practise safely without danger or cost
Slide	Single page inside presentation showing text and images together
Software	Set of programs that tells hardware what to do each time
Sort	Arranging data in order like A to Z or smallest largest
Sound	Music or voice played when program runs for effect
Spell Check	Tool that finds and suggests fixes for spelling mistakes instantly
Spreadsheet	Grid of rows and columns for organising numbers and data
Sprite	Character or object that moves and acts on the stage
Stage	Area where your characters move and show their actions live
Steps	Number of small moves sprite takes in chosen direction forward
Storage	Place where computer keeps data for later use safely
Story	Sequence of events told with words pictures and sound together
Subject	Short line telling what your email is about clearly
Subscribe	Following podcast to receive new episodes automatically when released
Symmetry	Both sides of design looking exactly same like mirror
System Software	Programs managing computer hardware and basic operations silently behind scenes
Tab	Small window letting you open many sites together
Technology	Tools and machines that make our daily work easier
Template	Ready-made design that keeps all slides looking consistent neatly
Test	Running program to check if it works as expected correctly
Theme	Set of colours and fonts giving uniform look to all slides
Timeline	Bar showing order and length of clips in your project
Tool	Option like brush or eraser that helps you create art
Trace	Following code line by line to find where mistake happens
Track	Single line holding one sound clip inside the project
Transistor	Small switch that replaced tubes and made computers smaller
Transition	Effect that moves you smoothly from one slide next
Trigger	Signal that causes sprite to begin doing something immediately afterwards
Trim	Cutting extra silence from start or end of clip
Tutorial	Step-by-step guide teaching you how to use a new tool
URL	Address that shows exact location of a website online
Update	Changing value already stored inside a variable to new value
Upgrade	Moving to better version with more powerful features than before
Upload	Sending your file from device to cloud storage
Vacuum Tube	Glass switch used in earliest large room-sized computers
Value	Number or word stored inside a variable for later use
Variable	Named box storing a value that can change during program
Version	Number showing which release of software you are using currently on device
Video	Moving pictures made from many still images played quickly
Video Call	Talking face-to-face through camera and internet at the same time
Voiceover	Recorded speech added over pictures to explain story clearly
Volume	How loud or soft the sound plays to listeners

Wizard	Series of screens helping you set up software without needing expert knowledge
Word Processor	Program for typing and formatting text documents beautifully and neatly

Self-Assessment

Chapter	I Can	With Help	Not Yet
Internet and ICT Environment	☐	☐	☐
Graphics in ICT	☐	☐	☐
Audio & Visual Communication	☐	☐	☐
Data Representation & Processing	☐	☐	☐
Programming	☐	☐	☐
Software Applications	☐	☐	☐